

ENTERPRISE WORKFORCE PRODUCTIVITY & EMPLOYEE ATTRITION ANALYTICS SYSTEM

End-to-End Data Analytics Project Report

PROJECT DETAIL	VALUE
Project ID	DA-W4-005
Role	Data Analyst Intern
Project Duration	Week 4 Internship Sprint
Tools	Python SQL Microsoft Excel Power BI
Reporting Scope	Workforce Productivity & Employee Attrition

Prepared as part of Data Analyst Internship Program

SOURCE-EVIDENCED • INTERNSHIP-SUBMISSION READY • PORTFOLIO READY

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1. Executive Summary

KPI	Verified Value	Management Meaning
Total Employees	4,980	Workforce population analysed.
Exited Employees	899	Workforce-level exited employee count.
Overall Attrition	18.05%	Share of the workforce represented by exits.
High + Critical Risk	955	Analytical early-warning population requiring prioritisation.

This project documents an end-to-end workforce analytics workflow completed as a Data Analyst Intern. The objective was to move from multiple HR datasets to a consolidated analytical view that management could use to understand workforce structure, attrition patterns, engagement, productivity and employee-level risk.

The workflow combined Python for profiling, cleaning, aggregation, exploratory analysis, correlation and modelling; SQL for business-oriented querying; Excel for formula-driven analysis, PivotTables, PivotCharts, slicers and dashboard construction; and Power BI for management-facing dashboards and risk exploration.

KEY FINDING

Overall attrition is 18.05%, with 899 exited employees. Detailed exit-reason, exit-type and tenure analysis uses 890 validated usable records; this is a validated analytical population difference, not a contradiction.

The evidence also shows that career growth, salary and work-life balance are the largest recorded exit reasons. Employee engagement dimensions show meaningful negative associations with attrition, while performance and overtime have near-zero linear correlation with attrition in the verified employee-level correlation output.

The predictive modelling exercise compared balanced Logistic Regression and Random Forest. Logistic Regression was tuned to a 0.70 threshold; Random Forest achieved higher accuracy but very low recall. The project therefore treats the models as analytical evidence rather than production-ready automated termination predictors.

2. Business Problem

The business problem was not simply to calculate an attrition percentage. HR information was distributed across employee, attendance, performance, engagement and attrition datasets, creating a need for consolidation, validation and interpretation.

- Create a consistent workforce-level view from multiple HR datasets.
- Identify where attrition is concentrated by department, age and salary band.
- Understand the main recorded reasons employees leave.
- Assess engagement, attendance, productivity and performance alongside attrition.
- Create an employee-level risk framework so HR can prioritise proactive review.
- Provide management-facing outputs through Excel and Power BI.
- Document real analytical errors, diagnosis and fixes rather than presenting only the final outputs.

WHY FOUR TOOLS?

Python provided reproducible analysis and modelling; SQL translated business questions into structured queries; Excel provided transparent operational reporting; Power BI provided interactive management dashboards.

3. Project Objectives

1. Clean HR datasets and validate keys, dates, ranges and relationships.
2. Analyse workforce structure and employee characteristics.
3. Measure attrition and identify major patterns.
4. Analyse salary, attendance, productivity, performance and engagement.

5. Build employee-level aggregated features.
6. Perform correlation analysis without treating correlation as causation.
7. Create a 0–100 employee risk score and classify risk levels.
8. Test Logistic Regression and Random Forest models.
9. Use SQL for joins, grouping, ranking, CASE, CTE and date-oriented analysis.
10. Develop Excel formulas, PivotTables, PivotCharts, slicers and a dashboard.
11. Develop six Power BI management pages.
12. Translate analytical outputs into practical HR recommendations.

4. Final Management Questions

Management Question	Evidence-Based Answer
Current attrition rate?	18.05% based on 4,980 employees and 899 exits.
Highest department attrition?	Engineering at 22% in the verified analysis.
Most affected age group?	20–24 at 31%, followed by 25–29 at 24%.
Does salary influence attrition?	Salary band is associated with attrition in the analysis; causation is not established.
Does overtime influence attrition?	Verified employee-level correlation with attrition = 0.005853, indicating a near-zero linear association.
Does engagement relate to retention?	Yes, engagement dimensions show negative associations with attrition; correlation does not prove causation.
Which departments have highest productivity?	Not conclusively established from the available analysis.
Which departments have attendance issues?	Not conclusively established from the available analysis.
Major exit reasons?	Career Growth 206; Salary 173; Work-Life Balance 160; Job Change 119; Management 88; Relocation 80; Personal Reasons 59; Unknown 5.
High-risk population?	955 employees are High + Critical Risk in the analytical risk framework.

5. Dataset Overview

Dataset	Purpose	Key / Relationship	Important Fields
Employees	Workforce master	Employee_ID; one row per employee	Department, Age, Experience, Salary, Employment_Status, Manager_ID, Joining_Date
Attendance	Daily/period attendance	Employee_ID; many records per employee	Date, Status, Work_Hours, Overtime_Hours, Late_Minutes, Remote_Work
Performance	Performance reviews	Employee_ID; many records per employee	Review_Date, Performance_Score, Productivity_Score, Quality_Score, Manager_Rating, Target_Achievement, Promotion_Eligible
Engagement	Employee survey measures	Employee_ID; repeated survey records	Survey_Date, Job_Satisfaction, Work-Life_Balance, Manager_Satisfaction, Career_Growth, Environment_Satisfaction, Engagement_Score
Attrition	Exited employee detail	Employee_ID; 0/1 relationship	Exit_Date, Exit_Reason, Exit_Type, Tenure_Months, Exit_Interview_Score

The Employee_ID is the primary relationship key. Repeated Attendance, Performance and Engagement records were therefore aggregated to employee level before being combined with the employee master for employee-level correlation, risk scoring and machine learning.

6. Data Architecture

The architecture is employee-centric. Employees acts as the master entity. Attendance, Performance and Engagement are one-to-many analytical histories, while Attrition is a zero-or-one relationship for an employee in the validated conceptual model. Manager_ID provides a self-reference for manager relationships.

Employee_ID	Employee_Name	Gender	Age	Department	Designation	Joining_Date	Experience_Year	Education	Employment_Type	Location	Manager_ID	Salary	Employment_Status	Attrition_Flag
E00001	Aditya Jadhav	Male	40	Procurement	Senior Procurement Specialist	28-01-2015	25	Bachelor's	Full-time	Pune	E02909	2696000	Active	0
E00002	Nikhil Bose	Female	41	Marketing	Marketing Manager	05-10-2025	21	Bachelor's	Full-time	Nashik	E04877	2542000	Active	0
E00003	Divya Mishra	Female	25	Operations	Operations Analyst	03-03-2020	3	Bachelor's	Full-time	Noida	E01498	668000	Exited	1
E00004	Pooja Reddy	Male	24	IT	Software Engineer	31-10-2023	0	Bachelor's	Full-time	Bengaluru	E04352	348000	Active	0
E00005	Pallavi Iyer	Male	36	Operations	Senior Operations Executive	30-12-2015	14	Bachelor's	Full-time	Noida	E02230	1828000	Exited	1
E00006	Anjali Bhosale	Female	29	Marketing	Digital Marketing Specialist	30-03-2021	8	MBA	Full-time	Pune	E03005	932000	Active	0
E00007	Pallavi Kamble	Male	38	Legal	Legal Analyst	08-11-2020	16	Master's	Full-time	Bengaluru	E04837	1778000	Active	0
E00008	Divya Shaikh	Female	27	Sales	Account Manager	18-04-2024	5	Master's	Full-time	Ahmedabad	E02820	1350000	Active	0
E00009	Rahul Banerjee	Female	34	HR	HR Business Partner	23-12-2017	10	Bachelor's	Full-time	Mumbai	E03089	956000	Active	0
E00010	Asha Sharma	Male	27	IT	Senior Data Analyst	03-12-2024	3	Master's	Full-time	Hyderabad	E01999	785000	Exited	1
E00011	Asha Singh	Female	34	Operations	Operations Executive	13-11-2022	13	Master's	Full-time	Bengaluru	E00624	1465000	Active	0
E00012	Kiran Patil	Male	41	Operations	Senior Operations Executive	09-02-2024	18	Bachelor's	Full-time	Bengaluru	E00624	1950000	Exited	1
E00013	Vishal Jadhav	Male	24	Marketing	Marketing Executive	20-12-2020	3	Master's	Full-time	Pune	E04070	493000	Exited	1
E00014	Neha Kulkarni	Male	42	Customer Support	Senior Support Executive	13-06-2022	20	Master's	Full-time	Delhi	E02495	2179000	Active	0
E00015	Vishal Shetty	Female	40	Sales	Sales Executive	07-05-2017	14	Bachelor's	Full-time	Mumbai	E03089	1456000	Active	0
E00016	Isha Khan	Female	35	HR	HR Analyst	29-08-2024	14	Diploma	Full-time	Bengaluru	E01781	1581000	Active	0
E00017	Ayush Reddy	Female	39	Marketing	Digital Marketing Specialist	03-01-2020	17	Bachelor's	Full-time	Gurugram	E04038	1892000	Active	0
E00018	Vijay Singh	Female	44	Operations	Senior Operations Executive	22-11-2018	19	Master's	Full-time	Chennai	E04046	2490000	Active	0
E00019	Pooja Reddy	Female	32	Sales	Business Development Executive	09-06-2024	6	Bachelor's	Full-time	Pune	E03489	794000	Active	0
E00020	Sneha Naik	Female	31	Engineering	Engineering Manager	18-01-2024	6	Bachelor's	Full-time	Chennai	E01655	1027000	Active	0
E00021	Arjun Kumar	Female	35	Marketing	Digital Marketing Specialist	23-12-2017	9	MBA	Full-time	Chennai	E00435	1350000	Active	0
E00022	Shreya Rao	Male	28	Operations	Senior Operations Executive	13-12-2021	6	Bachelor's	Full-time	Ahmedabad	E02463	1117000	Active	0
E00023	Aarav Singh	Female	39	IT	Senior Software Engineer	22-03-2019	16	Master's	Full-time	Mumbai	E02635	1710000	Active	0
E00024	Amit Kulkarni	Female	43	IT	IT Manager	14-06-2019	17	Bachelor's	Full-time	Mumbai	E02374	1917000	Active	0
E00025	Rakesh Joshi	Male	30	Sales	Senior Sales Executive	08-06-2016	9	Bachelor's	Full-time	Mumbai	E03063	1476000	Active	0
E00026	Ayush Bose	Female	35	Operations	Operations Manager	17-03-2020	11	Diploma	Full-time	Pune	E00386	1654000	Active	0
E00027	Yash Iyer	Female	27	Procurement	Procurement Manager	25-01-2023	3	Bachelor's	Full-time	Pune	E03017	947000	Exited	1
E00028	Riya Deshmukh	Female	35	Marketing	Marketing Analyst	11-10-2024	11	Master's	Full-time	Hyderabad	E02168	1282000	Active	0
E00029	Mohit Das	Male	29	Engineering	Engineering Analyst	17-08-2017	8	Bachelor's	Full-time	Mumbai	E02732	1140000	Exited	1
E00030	Amit More	Female	28	IT	Data Analyst	11-05-2023	5	Diploma	Full-time	Chennai	E00387	745000	Active	0

Figure 1. Employee master evidence used as the workforce anchor.

Relationship	Cardinality	Business Purpose
Employees → Attendance	1:M	Track work and attendance behaviour over time.
Employees → Performance	1:M	Track performance review history.
Employees → Engagement	1:M	Track engagement survey history.
Employees → Attrition	1:0/1	Capture exit information where an employee has exited.
Employees.Manager_ID → Employees.Employee_ID	Self-reference	Represent manager-to-employee structure.

ARCHITECTURE PRINCIPLE

The Employee_ID key must remain consistent across tables. Key integrity is therefore a core validation step, not merely a technical detail.

7. Data Cleaning & Validation

The cleaning stage focused on structural quality, referential integrity, business-rule validity and analytical usability. Null removal alone would not have been sufficient because HR records can be syntactically complete while still being logically invalid.

7.1 Employees Validation

Check	Why	Validation / Action	Evidence
Duplicate rows	Prevent repeated workforce records	Check duplicated rows and review records	Employees_Clean screenshot
Duplicate Employee_ID	Protect relationship key	Check uniqueness of Employee_ID	Employees_Clean screenshot
Missing Employee_ID / Department	Prevent unusable master records	Profile null counts and inspect missing fields	Data Cleaning / Missing Values evidence

Age / Experience validity	Prevent impossible workforce attributes	Apply range and consistency checks	Validation evidence
Future Joining_Date	Prevent temporal leakage	Compare Joining_Date with analysis context	Validation evidence
Salary validity	Prevent invalid compensation values	Check negative/zero or structurally invalid values	Data Cleaning evidence
Employment_Status / Manager_ID	Protect category and relationship logic	Validate allowed statuses and manager references	Validation evidence

Employee_ID	Employee_Name	Gender	Age	Department	Designation	Joining_Date	Experience_Year	Education	Employment_Type	Location	Manager_ID	Salary	Employment_Status	Attrition_Flag
E00001	Aditya Jadhav	Male	49	Procurement	Senior Procurement Specialist	28-01-2015	25	Bachelor's	Full-time	Pune	E02909	2696000	Active	0
E00002	Nikhil Bose	Female	41	Marketing	Marketing Manager	05-10-2025	21	Bachelor's	Full-time	Nashik	E04877	2542000	Active	0
E00003	Divya Mishra	Female	25	Operations	Operations Analyst	03-03-2020	3	Bachelor's	Full-time	Noida	E01498	668000	Exited	1
E00004	Pooja Reddy	Male	24	IT	Software Engineer	31-10-2023	0	Bachelor's	Full-time	Bengaluru	E04352	348000	Active	0
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E00006	Anjali Bhosale	Female	29	Marketing	Digital Marketing Specialist	30-03-2021	8	MBA	Full-time	Pune	E03005	932000	Active	0
E00007	Pallavi Kamble	Male	38	Legal	Legal Analyst	04-11-2020	16	Master's	Full-time	Bengaluru	E04837	1778000	Active	0
E00008	Divya Shaikh	Female	27	Sales	Account Manager	18-04-2024	5	Master's	Full-time	Ahmedabad	E02820	1350000	Active	0
E00009	Rahul Banerjee	Female	34	HR	HR Business Partner	23-12-2017	10	Bachelor's	Full-time	Mumbai	E03089	956000	Active	0
E00010	Asha Sharma	Male	27	IT	Senior Data Analyst	03-12-2024	3	Master's	Full-time	Hyderabad	E01999	785000	Exited	1
E00011	Asha Singh	Female	34	Operations	Operations Executive	13-11-2022	13	Master's	Full-time	Bengaluru	E00624	1465000	Active	0
E00012	Kiran Patil	Male	41	Operations	Senior Operations Executive	09-02-2024	18	Bachelor's	Full-time	Bengaluru	E00624	495000	Exited	1
E00013	Vishal Jadhav	Male	24	Marketing	Marketing Executive	20-12-2020	3	Master's	Full-time	Pune	E04070	493000	Exited	1
E00014	Neha Kulkarni	Male	42	Customer Support	Senior Support Executive	13-06-2022	20	Master's	Full-time	Delhi	E02495	2179000	Active	0
E00015	Vishal Shetty	Female	40	Sales	Sales Executive	07-05-2017	14	Bachelor's	Full-time	Mumbai	E01781	1456000	Active	0
E00016	Isha Khan	Female	35	HR	HR Analyst	29-08-2024	14	Diploma	Full-time	Bengaluru	E01781	1581000	Active	0
E00017	Ayush Reddy	Female	39	Marketing	Digital Marketing Specialist	03-01-2020	17	Bachelor's	Full-time	Gurgaon	E04038	1892000	Active	0
E00018	Vijay Singh	Female	44	Operations	Senior Operations Executive	22-11-2018	19	Master's	Full-time	Chennai	E04046	2490000	Active	0
E00019	Pooja Reddy	Female	32	Sales	Business Development Executive	09-06-2024	6	Bachelor's	Full-time	Pune	E03489	794000	Active	0
E00020	Sneha Naik	Female	31	Engineering	Engineering Manager	18-01-2024	6	Bachelor's	Full-time	Chennai	E01655	1027000	Active	0
E00021	Arjun Kumar	Female	35	Marketing	Digital Marketing Specialist	22-12-2017	9	MBA	Full-time	Chennai	E00435	1350000	Active	0
E00022	Shreya Rao	Male	28	Operations	Senior Operations Executive	13-12-2021	6	Bachelor's	Full-time	Ahmedabad	E02463	1117000	Active	0
E00023	Aarav Singh	Female	39	IT	Senior Software Engineer	22-03-2019	16	Master's	Full-time	Mumbai	E02635	1710000	Active	0
E00024	Amit Kulkarni	Female	43	IT	IT Manager	14-06-2019	17	Bachelor's	Full-time	Mumbai	E02374	1917000	Active	0
E00025	Rakesh Joshi	Male	30	Sales	Senior Sales Executive	08-06-2016	9	Bachelor's	Full-time	Mumbai	E03063	1476000	Active	0
E00026	Ayush Bose	Female	35	Operations	Operations Manager	17-03-2020	11	Diploma	Full-time	Pune	E00386	1654000	Active	0
E00027	Yash Iyer	Female	27	Procurement	Procurement Manager	25-01-2023	3	Bachelor's	Full-time	Pune	E03017	947000	Exited	1
E00028	Riya Deshmukh	Female	35	Marketing	Marketing Analyst	11-10-2024	11	Master's	Full-time	Hyderabad	E02168	1282000	Active	0
E00029	Mohit Das	Male	29	Engineering	Engineering Analyst	17-08-2017	8	Bachelor's	Full-time	Mumbai	E02732	1140000	Exited	1
E00030	Amit More	Female	28	IT	Data Analyst	11-05-2023	5	Diploma	Full-time	Chennai	E00387	745000	Active	0

Figure 2. Employees_Clean worksheet after validation-oriented preparation.

7.2 Attendance Validation

Check	Business Reason	Action
Duplicate Attendance_ID	Avoid duplicate event records	Check uniqueness
Duplicate Employee_ID + Date	Avoid duplicate daily records	Check composite uniqueness
Invalid Employee_ID / orphan records	Protect relationships	Cross-check against Employees
Future dates / before joining / after exit	Protect timeline integrity	Compare dates with employee lifecycle
Status vs Work_Hours	Detect logical contradictions	Review Present/Absent/Leave combinations
Negative/excessive hours or overtime	Prevent invalid workload measures	Apply range checks
Late_Minutes / Remote_Work	Validate operational fields	Check allowed ranges/categories

Attendance_ID	Employee_ID	Date	Status	Work_Hours	Overtime_Hours	Late_Minutes	Remote_Work
AT0000011	E00001	31-03-2025	Present	6.91	1.35	0	Yes
AT0000012	E00001	16-12-2025	Present	9.62	0	45	No
AT0000013	E00001	07-11-2025	Present	6.82	2.47	0	No
AT0000014	E00001	16-09-2025	Present	8.11	0	0	Yes
AT0000015	E00001	14-04-2025	Present	8.1	0	5	Yes
AT0000016	E00001	21-05-2025	Present	8.29	0	20	No
AT0000017	E00001	27-11-2025	Present	9.72	0	10	No
AT0000018	E00001	08-01-2025	Present	8.54	0	10	Yes
AT0000019	E00001	21-03-2025	Present	8.26	0	0	No
AT0000020	E00001	05-06-2025	Present	7.44	0	30	No
AT0000021	E00001	12-09-2025	Present	7.04	0	15	Yes
AT0000022	E00001	13-06-2025	Present	6.66	0	0	Yes
AT0000023	E00001	11-03-2025	Present	9.21	0	10	No
AT0000024	E00001	18-06-2025	Present	8.59	0	0	No
AT0000025	E00001	04-06-2025	Present	8.57	4.56	0	No
AT0000026	E00001	11-12-2025	Present	7.93	1.18	0	No
AT0000028	E00001	30-01-2025	Present	8.92	1.94	0	No
AT0000029	E00001	05-03-2025	Present	5.85	0	0	Yes
AT0000030	E00001	25-06-2025	Present	8.55	0	0	Yes
AT0000031	E00001	17-01-2025	Present	8.33	0	0	No
AT0000032	E00001	13-01-2025	Present	8.17	0	0	No
AT0000033	E00001	24-11-2025	Present	7.48	0	0	No
AT0000034	E00001	17-12-2025	Present	7.71	0	0	Yes
AT0000035	E00001	28-03-2025	Present	8.05	0	0	No
AT0000036	E00001	21-11-2025	Present	8.71	0	0	No

Figure 3. Attendance_Clean worksheet evidence.

7.3 Performance, Engagement & Attrition Validation

Dataset	Key Checks	Treatment / Interpretation
Performance	Review_ID duplicates; Employee_ID; future Review_Date; score ranges; missing Quality_Score	Validate IDs/dates/ranges; median-based imputation only where actually performed.
Engagement	Duplicate Employee_ID + Survey_Date; future dates; satisfaction/work-life/career/engagement ranges	Validate survey structure and apply median-based cleaning where actually performed.
Attrition	Missing/duplicate attrition; Employee_ID; Exit_Date; Exit_Reason; Exit_Type; Tenure; Exit_Interview_Score	Validate exit records and isolate usable records for detailed analysis.

Review_ID	Employee_ID	Review_Date	Performance_Score	Productivity_Score	Quality_Score	Manager_Rating	Target_Achievement	Promotion_Eligible
RV000009	E00009	22-12-2025	77.2	79.7	68	3.8	98.9	No
RV000010	E00010	20-12-2025	52.8	66.5	59.2	2.2	61.6	No
RV000011	E00011	17-12-2025	80.7	83.4	72.3	3.4	90.4	Yes
RV000012	E00012	16-12-2025	75.9	88.4	72.6	3.2	110.5	No
RV000013	E00013	27-12-2025	82.7	81	87.1	3.6	86.2	Yes
RV000014	E00014	21-12-2025	70.5	69.2	56.1	3.3	65.5	No
RV000015	E00015	30-12-2025	58	52.7	53.3	2.5	74.5	No
RV000016	E00016	31-12-2025	84	58.8	80.2	3.7	89.7	Yes
RV000017	E00017	27-12-2025	68.9	61.6	70	3.3	67.1	No
RV000018	E00018	28-12-2025	87	88.2	88.2	3.7	95.8	Yes
RV000019	E00019	28-12-2025	60.8	66.6	58.3	2.8	74.6	No
RV000020	E00020	16-12-2025	74.8	82.3	79.1	3.3	108.5	No
RV000021	E00021	18-12-2025	74.4	54.5	70.5	2.8	96.9	No
RV000022	E00022	23-12-2025	74.4	87	79.3	3.3	99.3	Yes
RV000023	E00023	23-12-2025	74.4	74.7	72.6	3	82.6	No
RV000024	E00024	21-12-2025	0	70.1	65.7	3.3	82	No
RV000025	E00025	27-12-2025	74.4	64.3	71.2	2.7	104.8	No
RV000026	E00026	17-12-2025	57.6	48.6	59.9	3	70	No
RV000027	E00027	31-12-2025	70.8	73.7	69.7	2.6	52.3	No
RV000028	E00028	15-12-2025	89.4	91.9	91.4	4.1	117.8	Yes
RV000029	E00029	17-12-2025	74.9	71.6	78.4	2.9	108.6	No
RV000030	E00030	17-12-2025	78.4	77.1	73.5	3.5	90.6	Yes
RV000031	E00031	21-12-2025	62.2	74.5	63.5	3.2	88.3	No
RV000032	E00032	22-12-2025	74.2	74.5	66.4	3.4	82.6	No
RV000033	E00033	20-12-2025	87.6	74.5	88.6	3.9	90.2	Yes

Figure 4. Performance_Clean worksheet evidence.

Survey_Date	Job_Satisfaction	Work_Life_Balance	Manager_Satisfaction	Career_Growth	Environment_Satisfaction	Engagement_Score
11-11-2025	69.4	90	88.4	87	93.7	86.9
15-11-2025	78.1	70.8	58.7	73.4	71.7	72.8
13-11-2025	79.8	71.9	65.8	64.7	80.3	69.7
27-11-2025	56.9	65.3	49.3	55.5	64	61.4
24-11-2025	38	59.6	57.4	60.7	36.2	49.3
02-11-2025	64.1	45.1	63	86.6	64.2	61.4
07-11-2025	70.3	83.7	75.7	84	67.9	78.4
17-11-2025	66.5	69	74.6	82.1	64	72.2
19-11-2025	69	79.8	84.3	94.9	84.6	87.1
03-11-2025	54.3	67.9	73	56.3	48.6	62.5
29-11-2025	73.3	77.8	74.2	73.5	61.9	77.3
20-11-2025	47	42.7	51.6	56.6	51.3	48.7
11-11-2025	74.7	92	86.8	78.9	82	71
17-11-2025	56.8	64.4	69.4	62.3	66.3	71
25-11-2025	82.9	69.8	45.9	88	36.7	71
01-11-2025	45.8	34.3	34	38.2	48.9	71
09-11-2025	64.7	57.9	54.3	62	69.9	68.2
17-11-2025	68.8	71.5	68.1	79.3	65.2	62.3
20-11-2025	55.8	61.7	65.7	73	97	92
22-11-2025	88.5	88	97.6	88.3	81.8	80.7
11-11-2025	86.1	80.9	75.1	69.9	55.6	56.5
22-11-2025	57.4	64	63	43.5	42.5	53.2
24-11-2025	71	48	49.9	62.2	82.4	70.9
24-11-2025	71	71.6	71.3	57.3	74.1	52.5
15-11-2025	71	46.3	46	58.8		

Figure 5. Employee_Engagement_Clean worksheet evidence.

Employee_ID	Exit_Date	Exit_Reason	Exit_Type	Tenure_Months	Exit_Interview_Score
E00977	10-08-2016	Career Growth	Voluntary	9	43
E03603	15-09-2016	Personal Reasons	Voluntary	9	42
E02526	20-09-2016	Work-Life Balance	Voluntary	9	66
E03695	25-09-2016	Salary	Voluntary	16	61
E00005	28-09-2016	Relocation	Voluntary	9	79
E04090	06-12-2016	Salary	Involuntary	10	72
E01644	28-12-2016	Salary	Voluntary	21	74
E02605	23-03-2017	Job Change	Involuntary	26	56
E02325	29-03-2017	Work-Life Balance	Involuntary	21	56
E03743	07-05-2017	Work-Life Balance	Voluntary	14	76
E04540	15-05-2017	Personal Reasons	Voluntary	21	49
E00814	12-07-2017	Management	Involuntary	14	59
E01134	23-07-2017	Work-Life Balance	Voluntary	18	76
E03521	29-07-2017	Personal Reasons	Voluntary	29	66
E02083	05-09-2017	Career Growth	Voluntary	17	55
E03120	08-09-2017	Unknown	Voluntary	18	60
E01611	10-09-2017	Unknown	Involuntary	22	59
E03115	12-09-2017	Unknown	Voluntary	17	61
E03283	14-09-2017	Unknown	Voluntary	14	68
E01660	02-10-2017	Unknown	Voluntary	29	82
E04941	03-11-2017	Salary	Involuntary	9	52
E02519	04-11-2017	Work-Life Balance	Voluntary	12	87
E04216	24-11-2017	Career Growth	Voluntary	31	39
E04068	16-12-2017	Salary	Voluntary	17	77
E01523	14-02-2018	Job Change	Voluntary	31	47

Figure 6. Attrition_Clean worksheet evidence.

Missing Value Check

I check each column for missing values. Missing values can affect analysis and therefore need to be identified before deciding how they should be handled.

```
[9]: employees.isnull().sum()
```

```
[9]: Employee_ID      5
     Employee_Name    0
     Gender          0
     Age            0
     Department      8
     Designation     0
     Joining_Date    0
     Experience_Years 0
     Education       0
     Employment_Type 0
     Location        0
     Manager_ID      0
     Salary          0
     Employment_Status 1
     dtype: int64
```

Duplicate Record Check

I check for duplicate records because duplicate employee records can result in incorrect employee counts and inaccurate HR metrics.

```
[10]: print("Duplicate full records:", employees.duplicated().sum())
```

```
Duplicate full records: 2
```

Figure 7. Missing values and duplicate validation evidence.

The screenshot shows a Jupyter Notebook interface with the title 'HR_Analytics_Data_Cleaning.ipynb'. The current cell is titled 'Missing Department Validation' and contains the following text: 'Department is an important business dimension used for workforce, performance, productivity, engagement, and attrition analysis. I first identify employees with missing Department values and inspect their available information before deciding how to handle the missing values.'

The code in the cell is:

```
[39]: missing_department = employees_clean[
      employees_clean["Department"].isna()
      ]
```

The output of the code is a table of employees with missing department values. The table has 13 columns: Employee_ID, Employee_Name, Gender, Age, Department, Designation, Joining_Date, Experience_Years, Education, Employment_Type, Location, Manager_ID, and a final column with values 2, 1, 1, 1, 1, 1, 1. The rows are numbered 510 through 517. The 'Department' column for all these rows is 'NaN'.

	Employee_ID	Employee_Name	Gender	Age	Department	Designation	Joining_Date	Experience_Years	Education	Employment_Type	Location	Manager_ID	
510	E00511	Aarav Gupta	Female	42	NaN	Customer Success Analyst	25-07-2021 00:00	20	MBA	Full-time	Hyderabad	E04823	2
511	E00512	Divya Mehta	Female	34	NaN	HR Manager	31-12-2016 00:00	13	Bachelor's	Full-time	Mumbai	E01208	1
512	E00513	Yash More	Male	34	NaN	Data Analyst	03-06-2025 00:00	9	Master's	Full-time	Noida	E00869	1
513	E00514	Vikas Deshmukh	Male	32	NaN	Support Manager	29-01-2022 00:00	10	Master's	Full-time	Mumbai	E01875	1
514	E00515	Arjun Joshi	Female	36	NaN	Software Engineer	19-11-2017 00:00	15	Bachelor's	Full-time	Pune	E00561	1
516	E00517	Sonal More	Male	38	NaN	Design Engineer	30-09-2025 00:00	17	MBA	Full-time	Pune	E03862	1
517	E00518	Vishal Shetty	Female	36	NaN	Software Engineer	23-11-2020 00:00	13	Bachelor's	Full-time	Nashik	E03450	1

Figure 8. Validation checks and resulting evidence.

Dataset	Validation Check	Result	Status
Employees	Missing Employee ID	0 PASS	
Employees	Duplicate Employee ID	0 PASS	
Employees	Invalid Age / Experience	0 PASS	
Employees	Invalid Salary	0 PASS	
Attendance	Duplicate Attendance Records	0 PASS	
Attendance	Invalid Employee IDs	0 PASS	
Attendance	Invalid Work Hours / Overtime	0 PASS	
Performance	Invalid Employee IDs	0 PASS	
Performance	Invalid Performance Scores	0 PASS	
Engagement	Invalid Employee IDs	0 PASS	
Engagement	Invalid Engagement Scores	0 PASS	
Attrition	Invalid Employee IDs	0 PASS	
Attrition	Invalid Exit Dates / Reasons	0 PASS	

Figure 9. Data cleaning worksheet evidence.

8. Exploratory Data Analysis

Python profiling was used before modelling to understand structure, distributions, missingness and duplicates. The project evidence includes loading, inspection, library preparation and validation stages.

```
[1]: import pandas as pd
import numpy as np
import os

print("Libraries imported successfully")

Libraries imported successfully
```

Figure 10. Python library/import setup evidence.

2. Loading Employee Master Dataset

Employees is the master dataset because Employee_ID will be used as the common key to connect attendance, performance, engagement, and attrition data.

```
[4]: employees = pd.read_csv(
    dataset_path + r"\Employees.csv"
)

print("Employees dataset loaded successfully")

Employees dataset loaded successfully
```

```
[5]: employees.head()
```

	Employee_ID	Employee_Name	Gender	Age	Department	Designation	Joining_Date	Experience_Years	Education	Employment_Type	Location	Manager_ID	Salary
0	E00001	Aditya Jadhav	Male	49	Procurement	Senior Procurement Specialist	28-01-2015 00:00	25	Bachelor's	Full-time	Pune	E02909	2696
1	E00002	Nikhil Bose	Female	41	Marketing	Marketing Manager	05-10-2025 00:00	21	Bachelor's	Full-time	Nashik	E04877	2542
2	E00003	Divya Mishra	Female	25	Operations	Operations Analyst	03-03-2020 00:00	3	Bachelor's	Full-time	Noida	E01498	668
3	E00004	Pooja Reddy	Male	24	IT	Software Engineer	31-10-2023 00:00	0	Bachelor's	Full-time	Bengaluru	E04352	348
4	E00005	Pallavi Iyer	Male	36	Operations	Senior Operations Executive	30-12-2015 00:00	14	Bachelor's	Full-time	Noida	E02230	1828

Figure 11. Dataset loading and initial inspection in Python.

Data Type Inspection

I inspect the data types and non-null values of each column before performing data cleaning. This helps identify columns that may require data type conversion or validation.

```
[8]: employees.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5008 entries, 0 to 5007
Data columns (total 14 columns):
#   Column              Non-Null Count  Dtype  
---  --
0   Employee_ID         5003 non-null   object 
1   Employee_Name       5008 non-null   object 
2   Gender              5008 non-null   object 
3   Age                 5008 non-null   int64  
4   Department          5000 non-null   object 
5   Designation         5008 non-null   object 
6   Joining_Date        5008 non-null   object 
7   Experience_Years    5008 non-null   int64  
8   Education            5008 non-null   object 
9   Employment_Type     5008 non-null   object 
10  Location            5008 non-null   object 
11  Manager_ID          5008 non-null   object 
12  Salary              5008 non-null   int64  
13  Employment_Status   5007 non-null   object 
dtypes: int64(3), object(11)
memory usage: 547.9+ KB
```

Figure 12. Data inspection output including profiling/structure checks.

Python Operation	Purpose	Output / Interpretation	Next Step
df.head()	Inspect first records	Confirm columns and sample values	Check data types and quality
df.sample()	View random records	Detect unusual records beyond first rows	Validate patterns
df.shape	Confirm size	Understand row/column scale	Plan processing
df.columns	Confirm field names	Prevent incorrect references	Build validation logic
df.describe()	Summarise numeric fields	View central tendency/range	Investigate outliers
df.isnull().sum()	Measure missingness	Identify fields requiring attention	Clean/impute where supported
df.duplicated().sum()	Measure exact duplicate rows	Identify structural duplication	Remove/inspect duplicates
df.value_counts()	Understand categories	See category frequency	Use in EDA

9. Workforce Overview



Figure 13. Power BI Executive HR Overview — management-facing workforce view.

The workforce overview provides the consolidated management context for headcount, salary, department distribution and workforce indicators. It is the starting point for moving from descriptive workforce information to attrition-focused investigation.

Measure	Verified / Evidenced Value	Interpretation
Total workforce	4,980	Base population for overall workforce and attrition rate.
Exited employees	899	Workforce-level exit population.
Active employees	4,081	Remaining workforce population represented in the project KPI layout.
Attrition rate	18.05%	Overall exit rate used throughout the management narrative.

10. Attrition Analysis

Attrition analysis was structured around department, age group, salary band, exit reason, exit type and tenure. The analysis distinguishes the workforce-level exit count (899) from the 890 usable records used for detailed exit-reason, exit-type and tenure analysis.

Dimension	Verified Result	Business Reading
Overall	18.05% attrition	A meaningful retention issue requiring targeted investigation.
Department	Engineering 22%; HR 21%; Finance 20%; IT 15%	Engineering has the highest verified departmental rate.
Age	20–24 = 31%; 25–29 = 24%	Early-career employees show the highest observed attrition.
Salary	Lowest salary band = 35%; next groups ≈22%, ≈15%, ≈12%	Lower salary band is associated with higher observed attrition.
Exit reasons	Career Growth 206; Salary 173; WLB 160; Job Change 119; Management 88; Relocation 80; Personal 59; Unknown 5	Career development, compensation and work-life balance are the leading recorded reasons.
Exit type	Voluntary 732; Involuntary 153; Unknown 5	Voluntary exits dominate the detailed usable records.
Tenure	0–11 months 187; 12–23 months 201	388 exits occurred within first 24 months, ≈43.6% of 890.



Figure 14. Power BI Attrition Analytics dashboard.

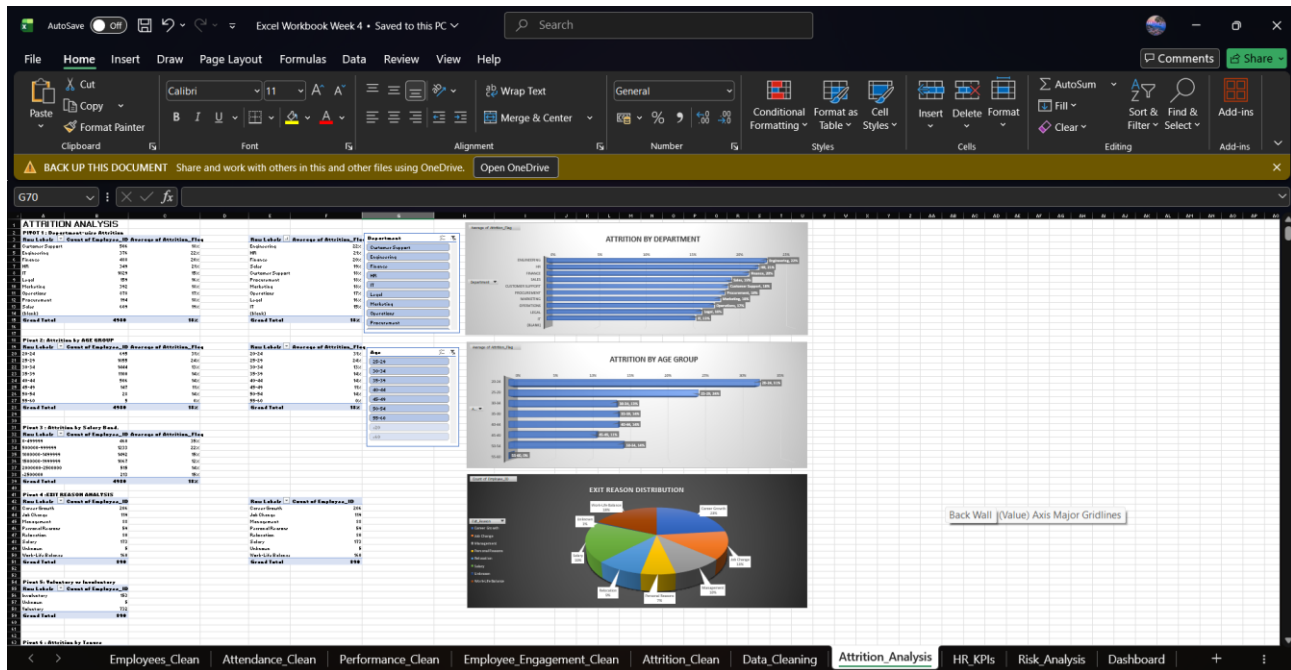


Figure 15. Excel Attrition_Analysis worksheet and chart evidence.

11. Salary Analysis

Salary was analysed as a workforce segmentation variable and as an attrition-related factor. The report uses association language because the available analysis does not establish causal impact.

INTERPRETATION

The analysis shows an association between salary band and attrition. It does not establish that salary alone causes an employee to leave.

Question	Evidence	Interpretation
Which salary group has highest attrition?	Lowest salary band = 35%	Lower compensation band shows the highest observed attrition.
Is salary related to attrition?	Correlation = -0.092928	Weak negative linear association in employee-level correlation.
What should HR do?	Review compensation and progression	Use salary findings alongside career growth and engagement evidence rather than in isolation.

12. Attendance Analysis

Attendance analysis considered Work_Hours, Overtime_Hours, Late_Minutes, Remote_Work and Status. These measures were aggregated to employee-level averages where required for combined analysis.

Metric	Use in Analysis	Verified Relationship
Avg Work Hours	Workload exposure	Correlation with attrition = 0.018315
Avg Overtime Hours	Additional workload	Correlation with attrition = 0.005853
Late Minutes	Attendance discipline / lateness	Used as a potential risk feature where supported
Remote Work	Work arrangement context	Used as a contextual employee feature
Status	Presence/absence/leave classification	Validated against Work_Hours logic

EVIDENCE BOUNDARY

A department-level attendance ranking is not conclusively established from the available analysis and is therefore not

presented as a fact.

13. Productivity Analysis

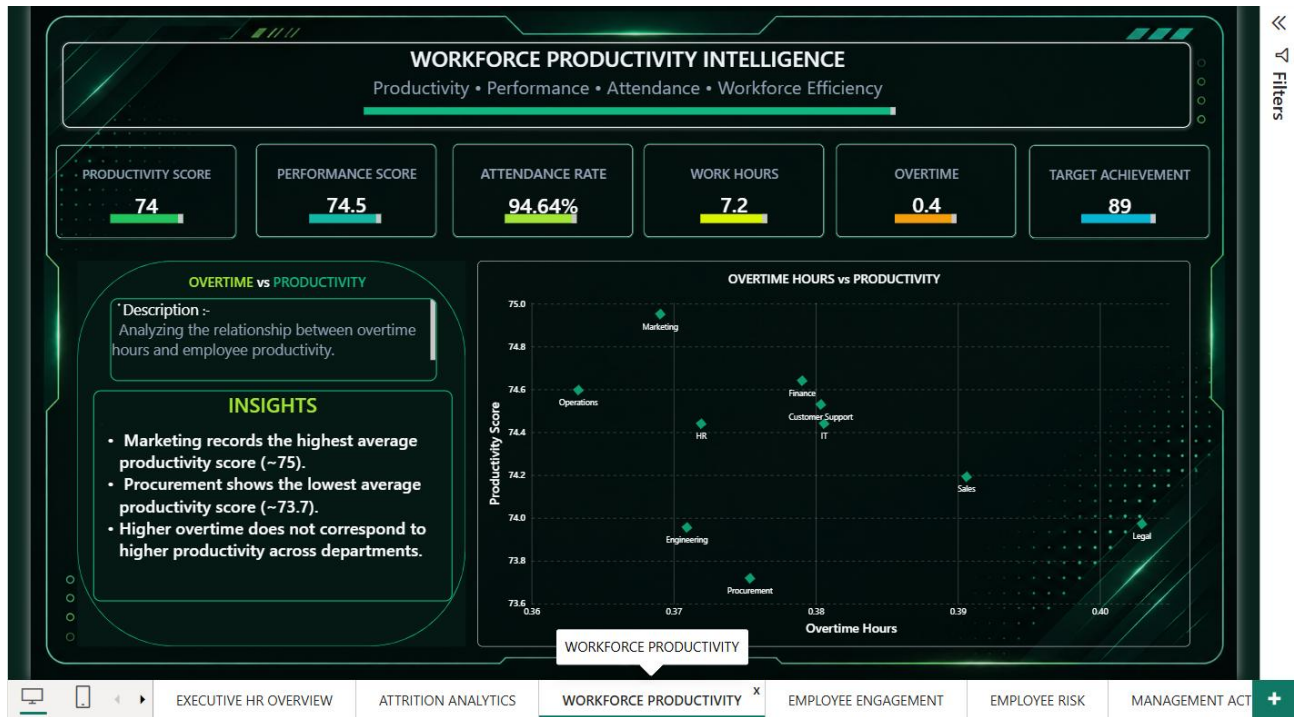


Figure 16. Power BI Workforce Productivity dashboard.

Productivity was evaluated using Productivity_Score alongside work and performance measures. The strongest verified internal relationship is Performance $\leftrightarrow$ Productivity = 0.759140, indicating a strong positive linear association in the employee-level analytical dataset.

Measure	Purpose	Verified Finding
Productivity_Score	Measure employee productivity	Core productivity outcome.
Performance_Score	Measure performance	Strong positive association with productivity: 0.759140.
Work_Hours / Overtime	Contextual workload measures	Near-zero linear attrition relationships in the verified correlation output.
Attendance	Operational context	Useful for workforce monitoring, but department ranking not conclusively established.

14. Performance Analysis

Measure	Business Purpose	Verified Evidence
Performance_Score	Overall performance	Correlation with attrition = -0.003718.
Productivity_Score	Output/productivity	Correlation with attrition = 0.004357.
Quality_Score	Quality of work	Validated as a performance field; missing values handled where actually performed.
Manager_Rating	Manager assessment	Included in analytical feature set.
Target_Achievement	Goal attainment	Included in performance interpretation.
Promotion_Eligible	Career movement indicator	Relevant to retention/career-growth analysis.

The near-zero attrition correlations for performance and productivity should not be interpreted as proof that performance is irrelevant to retention. They indicate that a simple linear relationship with attrition was weak in this dataset.

15. Employee Engagement Analysis



Figure 17. Power BI Employee Engagement dashboard.

Engagement was analysed through Job Satisfaction, Work-Life Balance, Manager Satisfaction, Career Growth, Environment Satisfaction and Engagement Score. These measures are especially relevant because the strongest negative attrition associations in the verified correlation output come from engagement-related dimensions.

Dimension	Attrition Correlation	Internal Engagement Relationship
Job Satisfaction	-0.223238	Job Satisfaction ↔ Engagement = 0.818527
Work-Life Balance	-0.239754	Work-Life Balance ↔ Engagement = 0.804877
Manager Satisfaction	-0.225597	Manager Satisfaction ↔ Engagement = 0.817671
Career Growth	-0.207363	Career Growth ↔ Engagement = 0.796415
Engagement Score	-0.262206	Overall engagement indicator.

BUSINESS MEANING

Lower engagement-related scores are associated with higher attrition in the employee-level analysis. This supports targeted retention interventions, while correlation alone does not prove causation.

16. Correlation Analysis

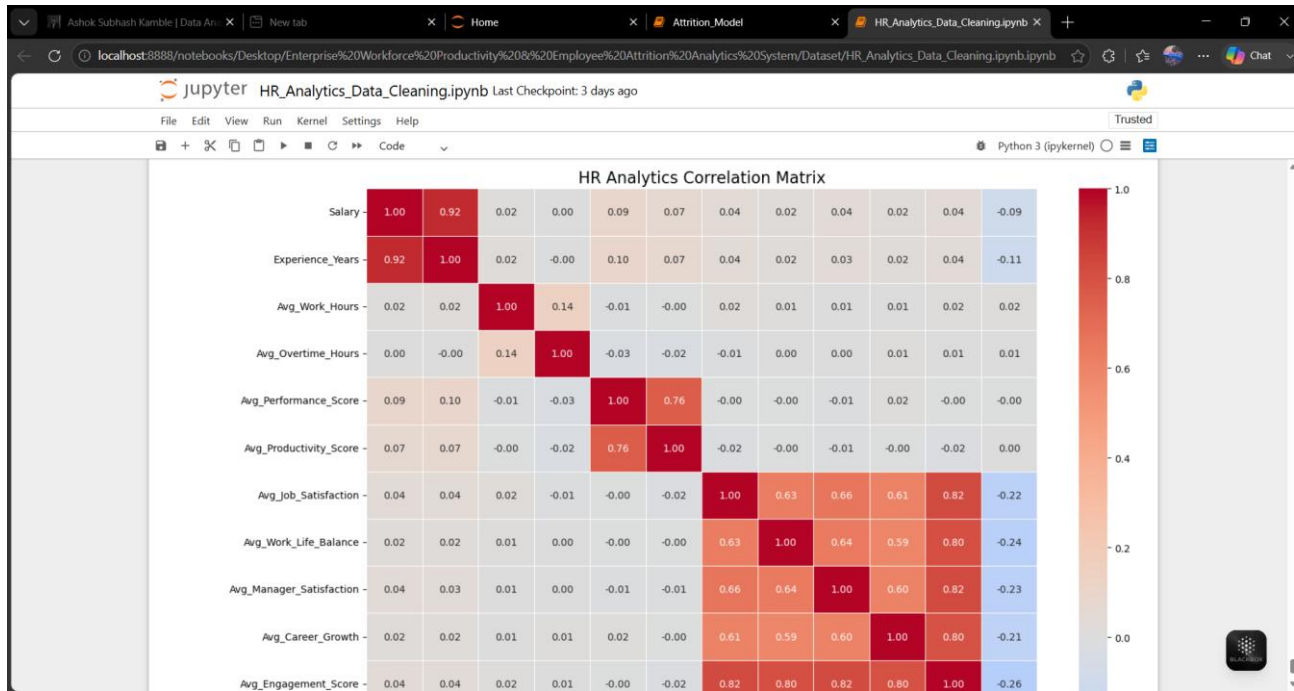


Figure 18. Employee-level correlation heatmap evidence.

Variable	Correlation with Attrition
Salary	-0.092928
Experience_Years	-0.110673
Avg_Work_Hours	0.018315
Avg_Overtime_Hours	0.005853
Avg_Performance_Score	-0.003718
Avg_Productivity_Score	0.004357
Avg_Job_Satisfaction	-0.223238
Avg_Work_Life_Balance	-0.239754
Avg_Manager_Satisfaction	-0.225597
Avg_Career_Growth	-0.207363
Avg_Engagement_Score	-0.262206

Internal Relationship	Correlation
Job Satisfaction ↔ Engagement	0.818527
Work-Life Balance ↔ Engagement	0.804877
Manager Satisfaction ↔ Engagement	0.817671
Career Growth ↔ Engagement	0.796415
Performance ↔ Productivity	0.759140

IMPORTANT NOTE

Correlation measures linear association. It does not establish causation, direction of causality, or whether an intervention will definitely change attrition.

17. Employee Risk Scoring

```
[ ]: risk_order = [
    "LOW RISK",
    "MEDIUM RISK",
    "HIGH RISK",
    "CRITICAL"
]

risk_attrition_analysis["Risk_Category"] = pd.Categorical(
    risk_attrition_analysis["Risk_Category"],
    categories=risk_order,
    ordered=True
)

risk_attrition_analysis = (
    risk_attrition_analysis
    .sort_values("Risk_Category")
    .reset_index(drop=True)
)

risk_attrition_analysis

[ ]: print(
    "Total Employees:",
    risk_attrition_analysis["Total_Employees"].sum()
)

print(
    "Total Exited:",
    risk_attrition_analysis["Exited_Employees"].sum()
)
```

Figure 19. Risk-scoring implementation evidence in Python.



Figure 20. Power BI Employee Risk dashboard.

Risk Score	Level	Employees
0-30	Low Risk	1,182
31-60	Medium Risk	2,843
61-80	High Risk	905
81-100	Critical Risk	50
61-100	High + Critical	955

The High + Critical population is 955 employees, approximately 19.2% of the workforce. The risk framework can incorporate supported factors such as low job satisfaction, low salary, high overtime, poor performance, low career growth, recent joining and high absenteeism.

RISK INTERPRETATION

Risk scoring is an analytical early-warning framework and not a guaranteed prediction of employee exit.

Risk Level	HR Action
Critical	Individual intervention and retention review.
High	Prioritised manager/HR review.
Medium	Preventive engagement and monitoring.
Low	Routine monitoring.

18. Power BI Risk Export → Excel

The project included a practical workflow for taking the Power BI employee-risk detail into Excel for operational use.

Step	Click / Action	Expected Output	Evidence
1	Open Employee Risk page	Risk dashboard displayed	Employee Risk screenshot
2	Select Employee Risk Details	Detailed table becomes active	Risk dashboard evidence
3	Click three dots	Visual menu opens	Export workflow
4	Select Export data	Export dialog appears	Export workflow
5	Save CSV	CSV file created	Excel workflow
6	Open CSV in Excel	Rows become available	XLOOKUP evidence
7	Inspect columns	Identify missing Employee_ID	XLOOKUP screenshot
8	Use XLOOKUP	Employee_ID recovered from master	XLOOKUP screenshot
9	Validate	Spot-check IDs / Not Found cases	XLOOKUP evidence

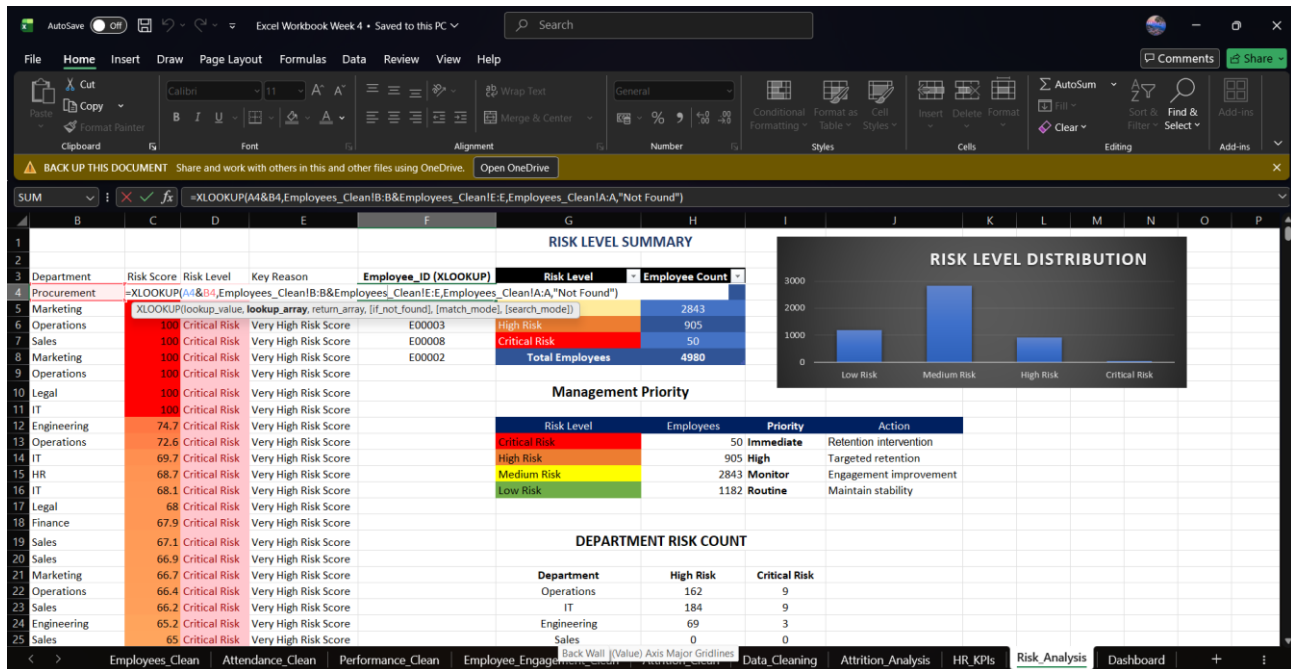


Figure 21. Power BI-to-Excel risk export and XLOOKUP recovery evidence.

19. Excel Workbook Development

Sheet	Purpose	Business Role
Employee_Master	Validated workforce master	Primary employee reference.
Attendance	Attendance history	Workload and attendance analysis.
Performance	Performance history	Performance/productivity analysis.
Engagement	Survey history	Engagement and retention analysis.
Attrition	Exit details	Attrition and exit-reason analysis.
Data_Cleaning	Validation support	Document data-quality checks.
HR_KPIs	KPI calculations	Management summary.
Attrition_Analysis	Pivot/chart analysis	Department/age/salary/exit analysis.

Risk_Analysis	Employee risk detail	Risk prioritisation and formulas.
Dashboard	Management dashboard	Consolidated Excel reporting.

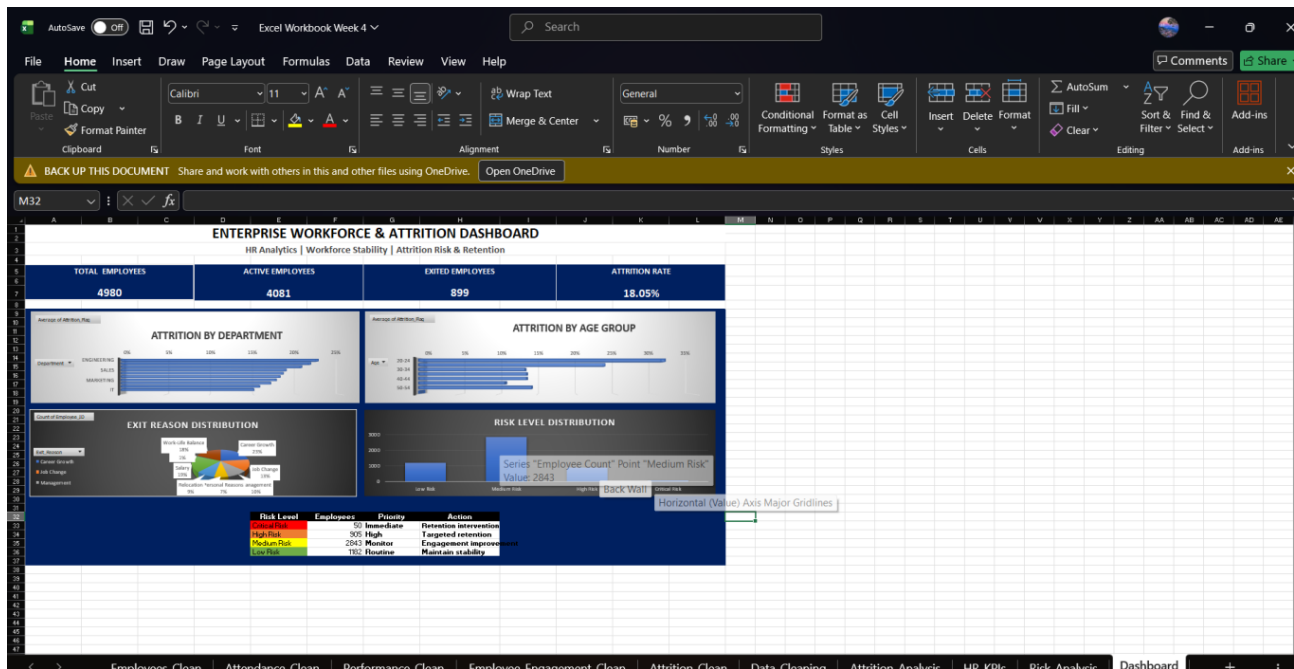


Figure 22. Excel Dashboard development evidence.

20. Excel Formula Implementation

20.1 XLOOKUP

```
=XLOOKUP(A4&B4,Employees_Clean!B:B&Employees_Clean!E:E,Employees_Clean!A:A,"Not Found")
```

Component	Meaning
A4	Employee_Name in the risk export.
B4	Department in the risk export.
Employees_Clean!B:B	Employee_Name lookup field.
Employees_Clean!E:E	Department lookup field.
Employees_Clean!A:A	Employee_ID returned.
"Not Found"	Fallback when no matching record exists.

Employee_ID was missing from the exported risk detail. Name + Department was therefore used as a composite lookup key. This is useful for recovery but has an inherent limitation: names and departments may not be unique.

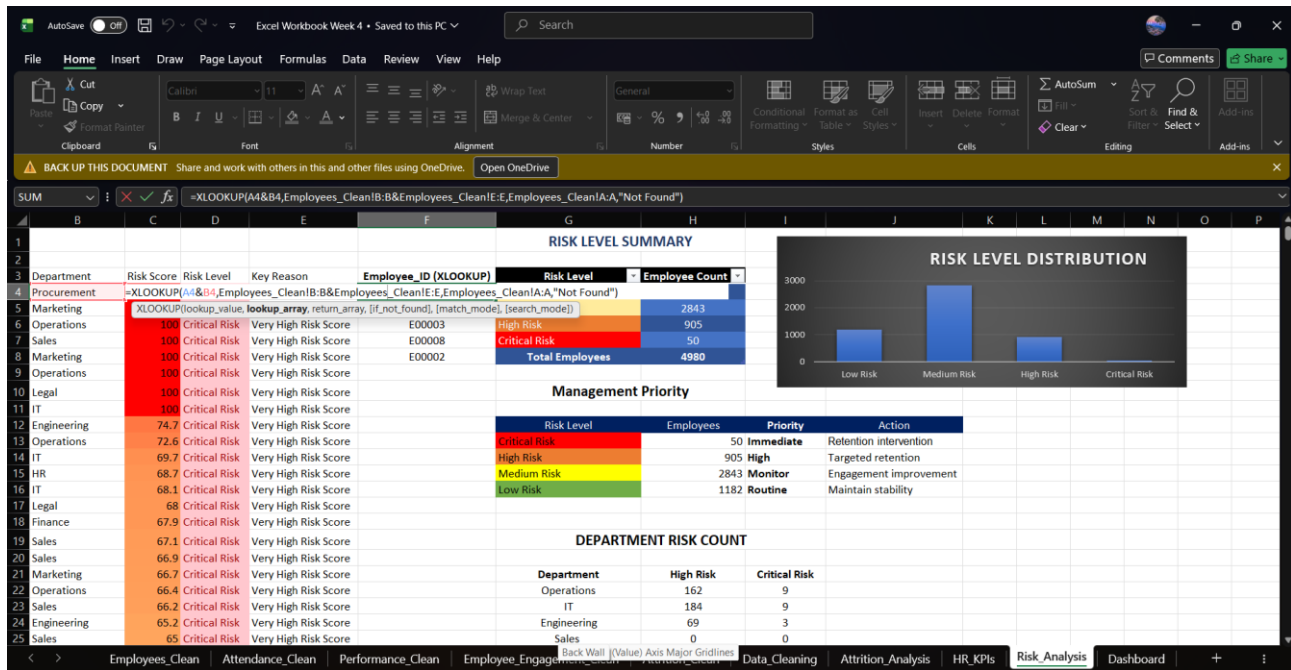


Figure 23. XLOOKUP formula and populated Employee_ID evidence.

20.2 COUNTIFS

```
=COUNTIFS($B:$B,G22,$D:$D,H$21)
```

Purpose: count employees by Department × Risk Level. The row criterion is the department and the column header supplies the risk level, allowing the table to be filled across and down.

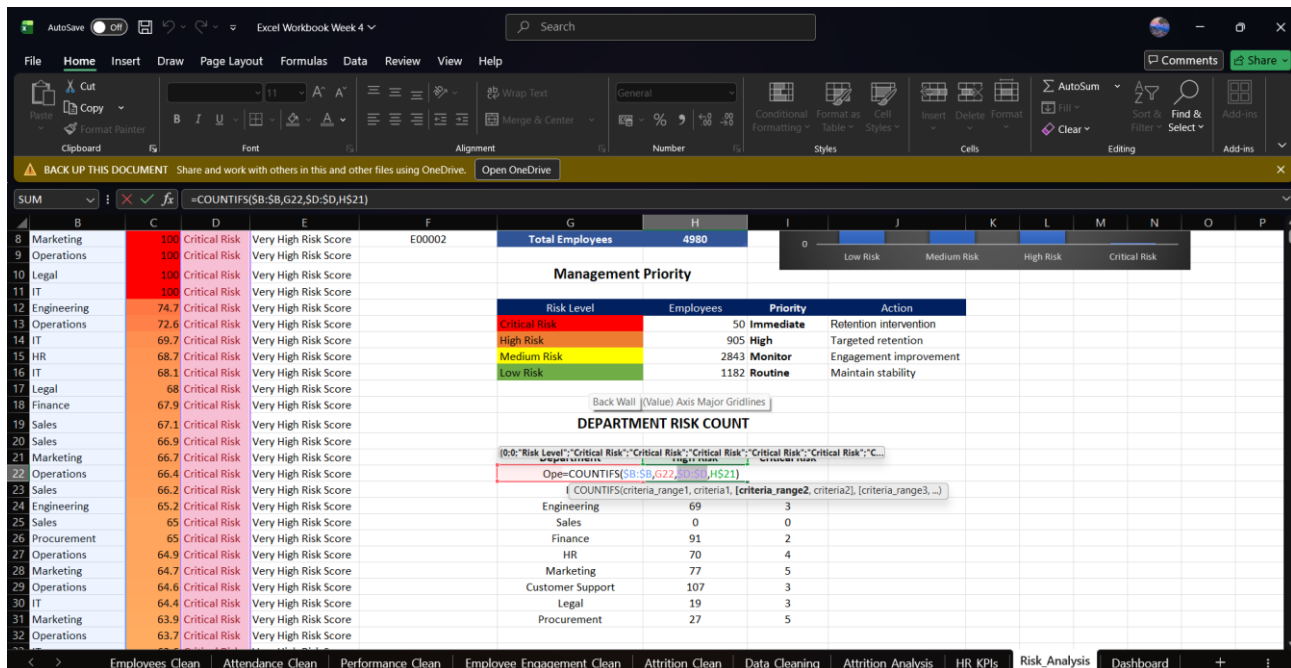


Figure 24. COUNTIFS-based department risk count evidence.

20.3 SUMIFS

```
=SUMIFS($C:$C,$B:$B,G37)
```

Purpose: calculate total Risk Score by Department. The formula sums the Risk Score column for rows matching the department in G37.

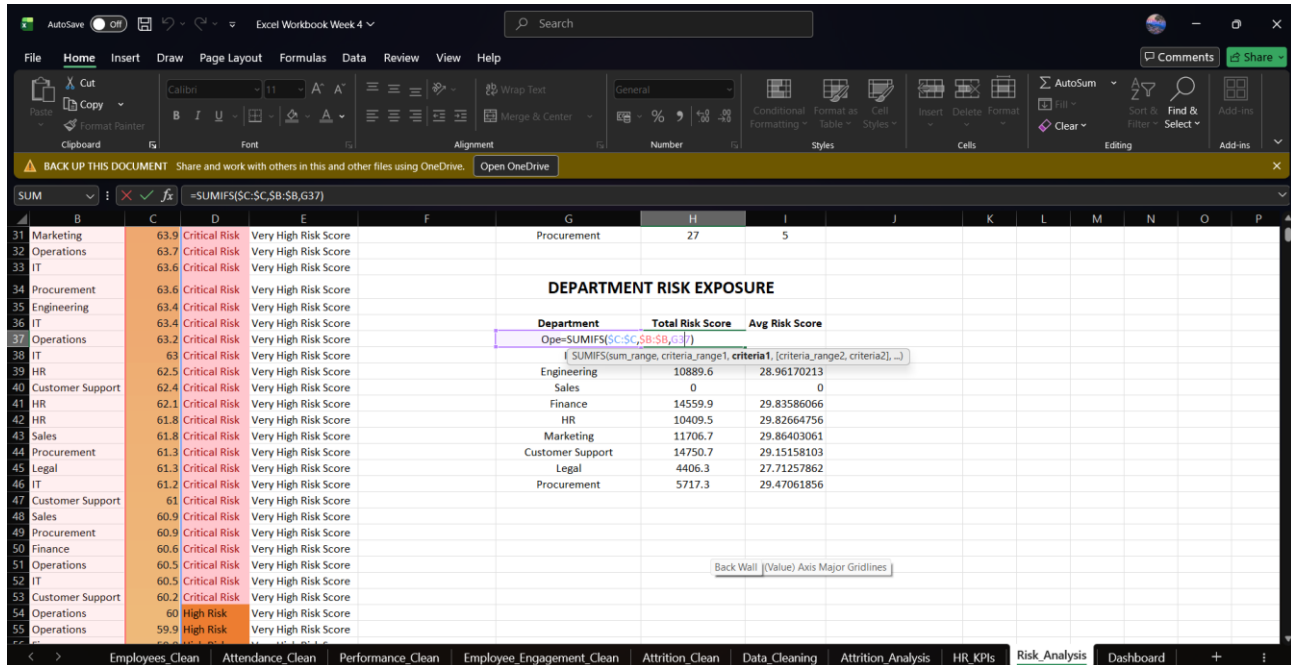


Figure 25. SUMIFS-based department risk exposure evidence.

20.4 AVERAGEIFS + IFERROR

`=IFERROR(AVERAGEIFS($C:$C,$B:$B,G37),0)`

The original department average returned #DIV/0! for Sales because the matching set contained no usable numerical values for the calculation. IFERROR converts the error to a controlled 0 result and prevents a broken dashboard calculation.

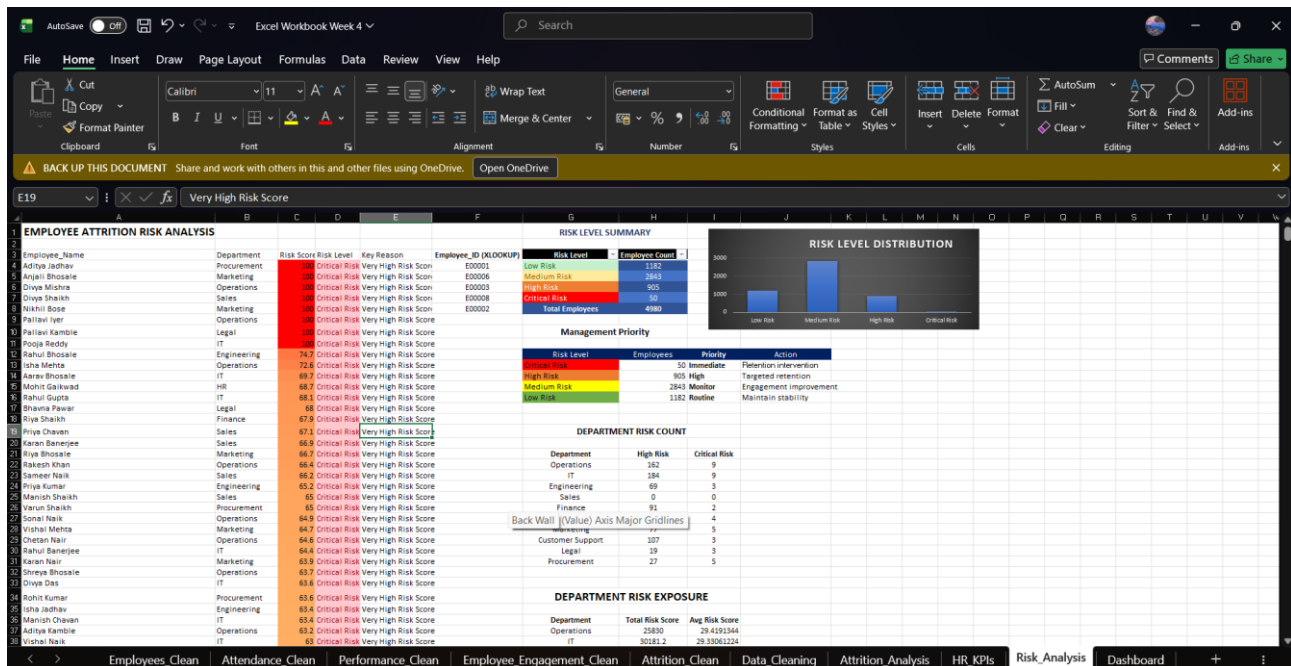


Figure 26. Excel calculation area showing the department risk analysis workflow.

ERROR FIX

The diagnostic sequence was: observe #DIV/0! → inspect matching records → determine that the match set had no usable

numerical values → wrap AVERAGEIFS in IFERROR → verify the corrected output.

20.5 IF — Attrition Flag

=IF(Employee_Master!N2="Exited",1,0)

Exited is encoded as 1 and Active as 0. This binary flag makes attrition usable in aggregation, correlation and modelling workflows.

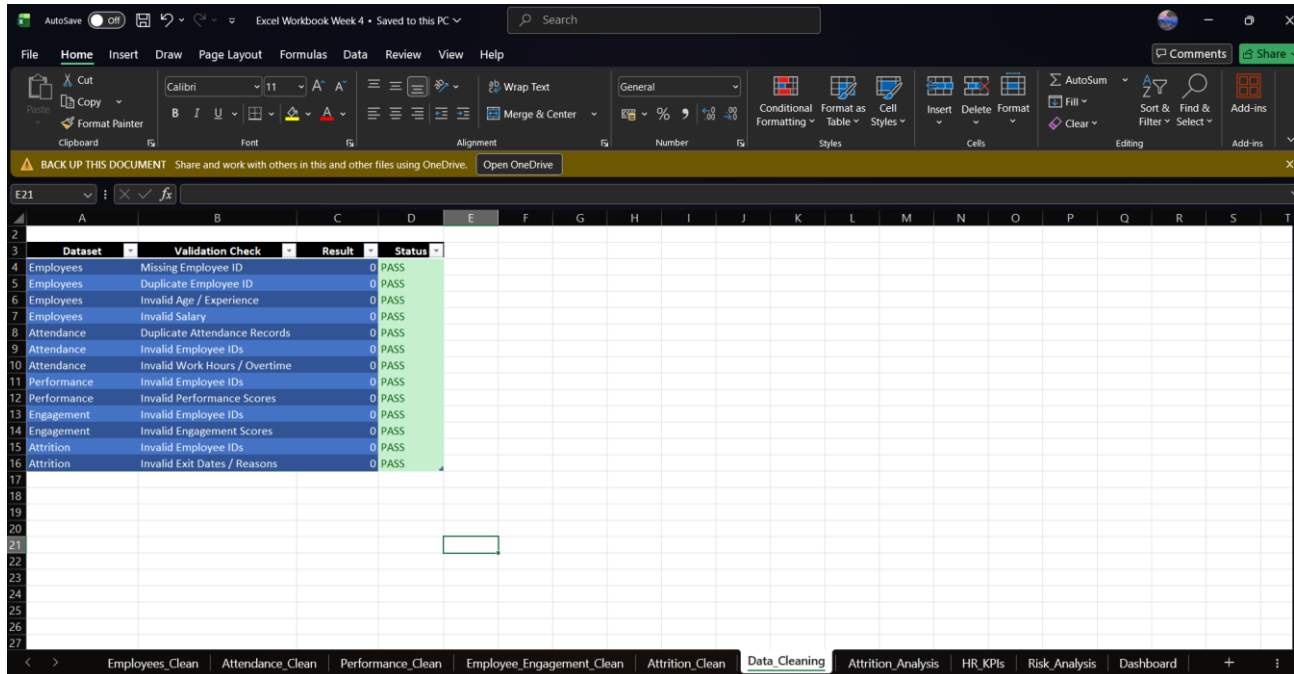


Figure 27. Excel cleaning/calculation evidence supporting structured analytical fields.

21. PivotTables

PivotTable	Business Question	Rows	Values
Department-wise Attrition	Where is attrition concentrated?	Department	Count of Employee_ID / attrition measure
Age Group Attrition	Which age groups have higher attrition?	Age Group	Count of exits
Salary Band Attrition	Which salary band shows highest attrition?	Salary Band	Count / rate
Exit Reason	Why are employees leaving?	Exit Reason	Count of exits
Voluntary vs Involuntary	What type of exits dominate?	Exit Type	Count of exits
Tenure	When do exits occur?	Tenure Band	Count of exits
Gender	How does attrition vary by gender?	Gender	Count of exits

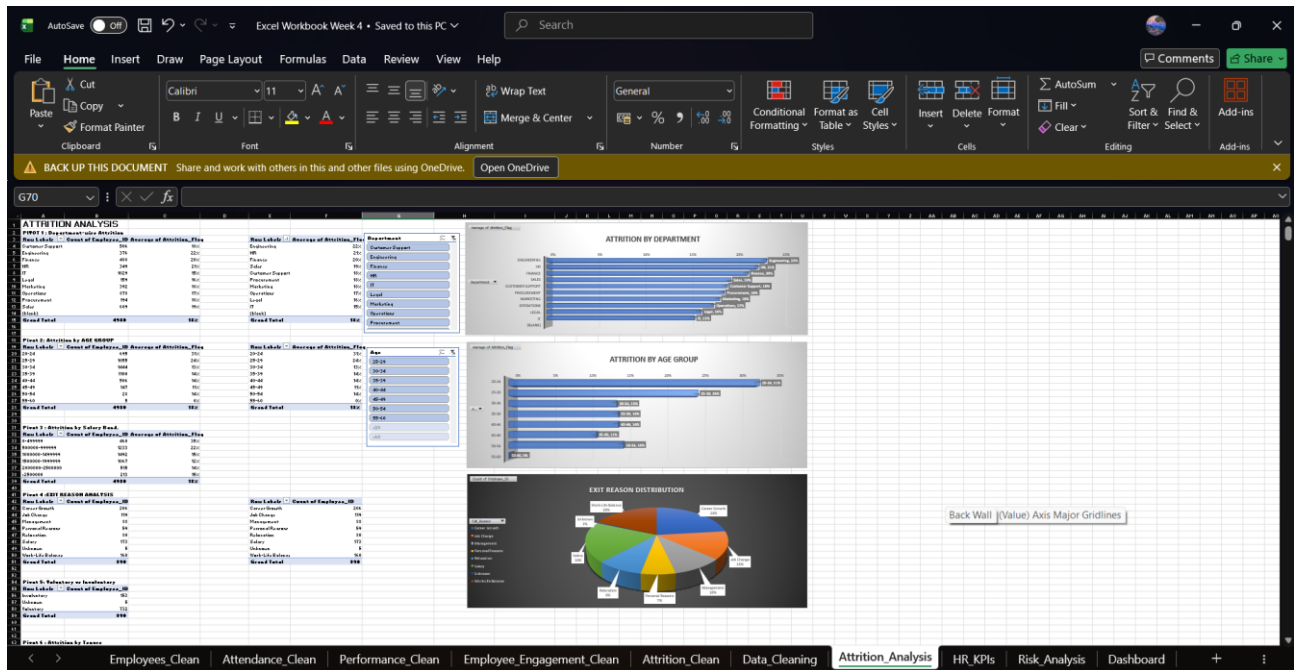


Figure 28. PivotTable/PivotChart-style attrition analysis evidence.

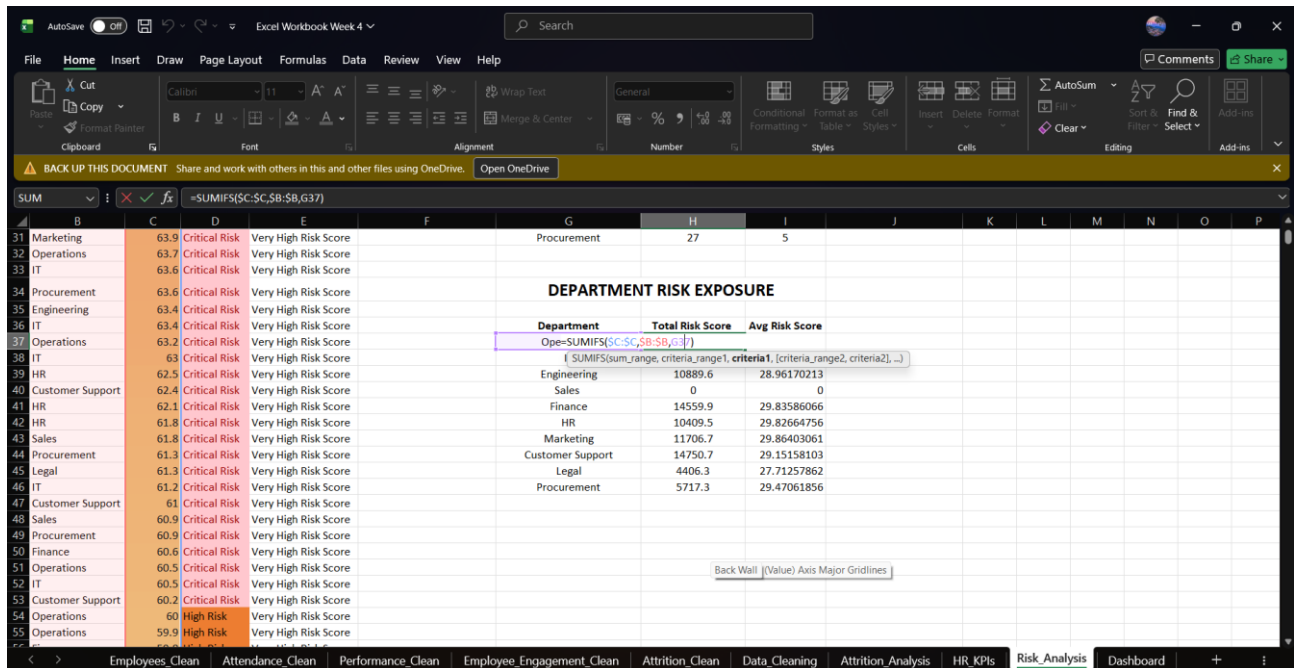


Figure 29. Pivot/summary calculation evidence used alongside the analysis.

22. PivotCharts

Chart	Type	Purpose
Attrition by Department	Horizontal Bar	Compare department attrition.
Attrition by Age Group	Horizontal Bar	Highlight early-career patterns.
Exit Reason Distribution	Pie	Show reason mix.
Risk Level Summary	Chart / KPI view	Use only where final evidence confirms completion.

PIVOTCHART PROBLEM

Removing Count of Employee_ID from the PivotChart also changed the connected PivotTable because the PivotChart and

PivotTable share the same pivot field structure. A copied/static summary table can be used when an independent dashboard chart is required.

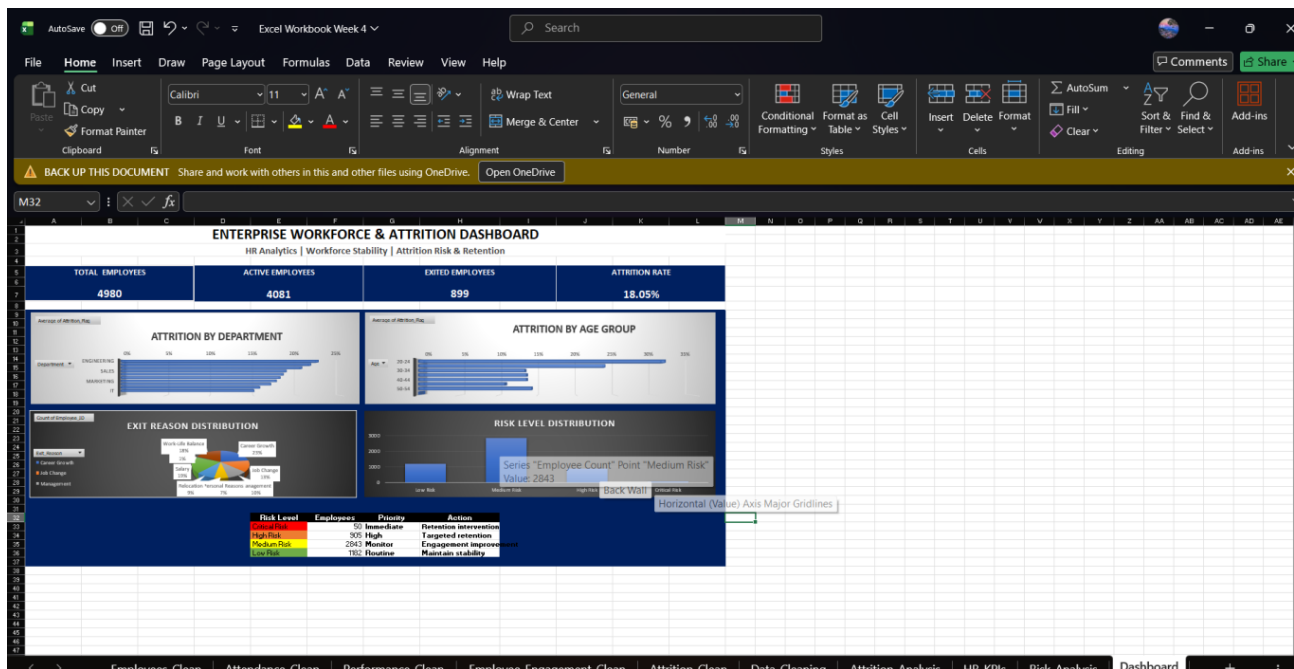


Figure 30. Excel dashboard/chart composition evidence.

23. Conditional Formatting

Field	Rule	Visual Meaning
Risk Level	Low Risk → Green	Lower priority.
Risk Level	Medium Risk → Yellow	Preventive attention.
Risk Level	High Risk → Orange	Priority review.
Risk Level	Critical Risk → Red	Immediate HR attention.
Risk Score	Green → Yellow → Red	Continuous visual risk scale.

Conditional formatting turns the risk table into an operational prioritisation tool: HR can scan the sheet and immediately identify employees requiring attention.

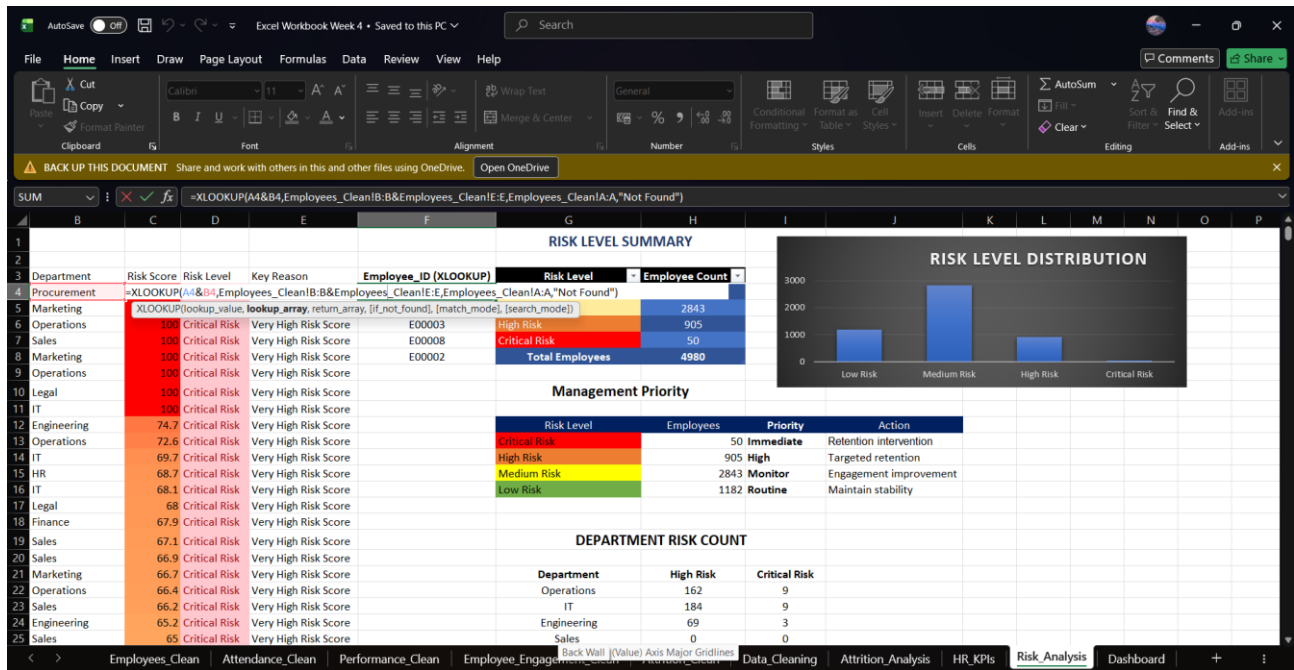


Figure 31. Risk analysis worksheet with conditional formatting evidence.

24. Slicers

Step	Excel Click Path	Result
1	Select PivotTable	PivotTable becomes active.
2	PivotTable Analyze → Insert Slicer	Slicer selection dialog opens.
3	Select Department	Department slicer created.
4	Select Age	Age slicer created.
5	OK → resize → position	Slicers placed on dashboard.
6	Report Connections	Connect relevant PivotTables/charts.
7	Test filtering	Confirm dashboard responds to selections.

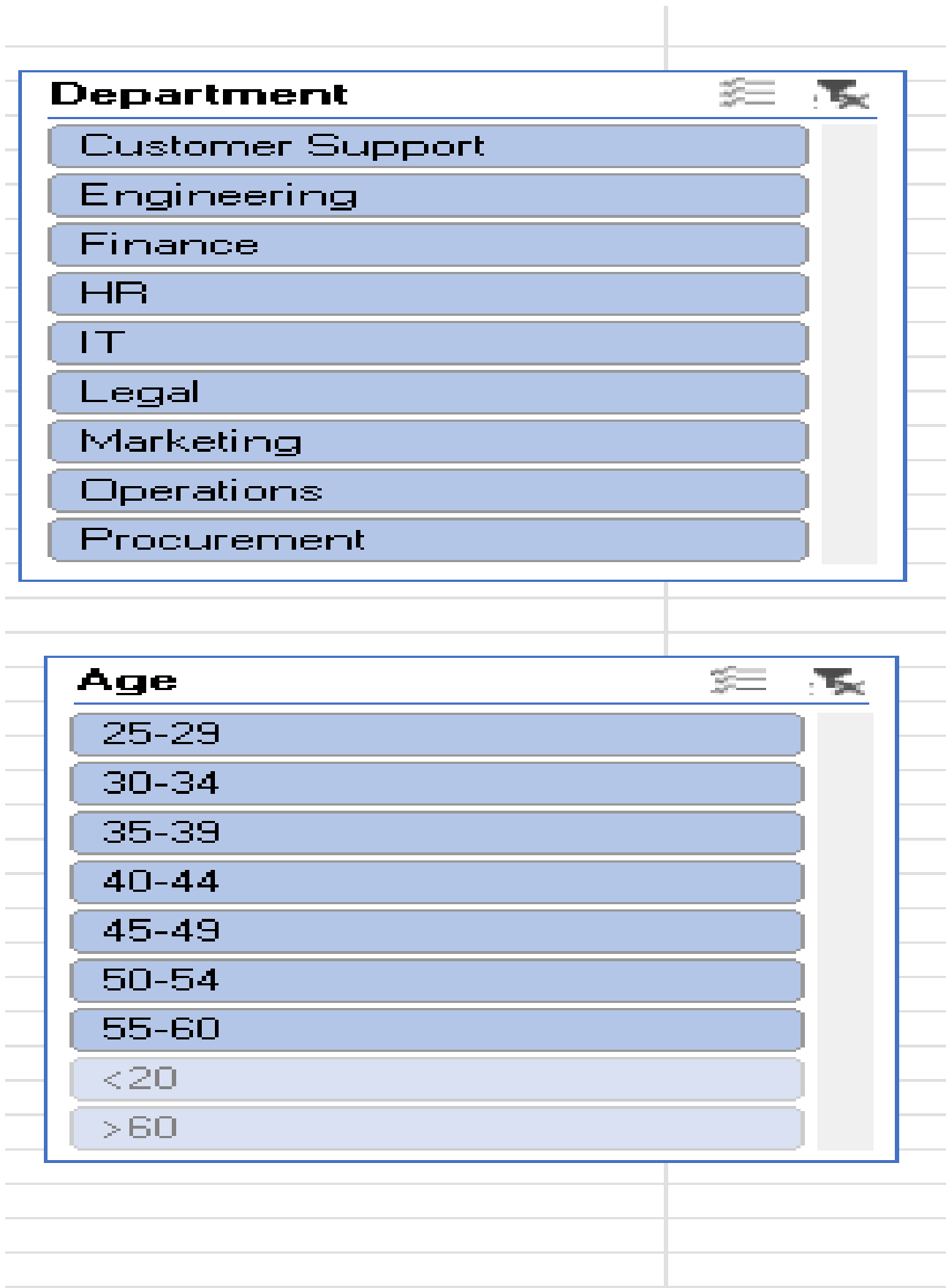


Figure 32. Department and Age slicer evidence.

SLICER ISSUE

A duplicate Department slicer was accidentally created. It was identified by reviewing the dashboard and Report Connections, then the duplicate was deleted and the final slicer configuration was verified.

25. Excel Dashboard Development

The Excel dashboard evolved from an initial layout that looked too small into a broader A:L structure. The final layout uses a title, subtitle, KPI cards, charts, slicers and risk summary elements.

Dashboard Area	Final Layout / Content
Total Employees	A5:C8 — 4,980
Active Employees	D5:F8 — 4,081
Exited Employees	G5:I8 — 899
Attrition Rate	J5:L8 — 18.05%
Supporting visuals	Attrition, risk and management-oriented charts
Filters	Department and Age slicers

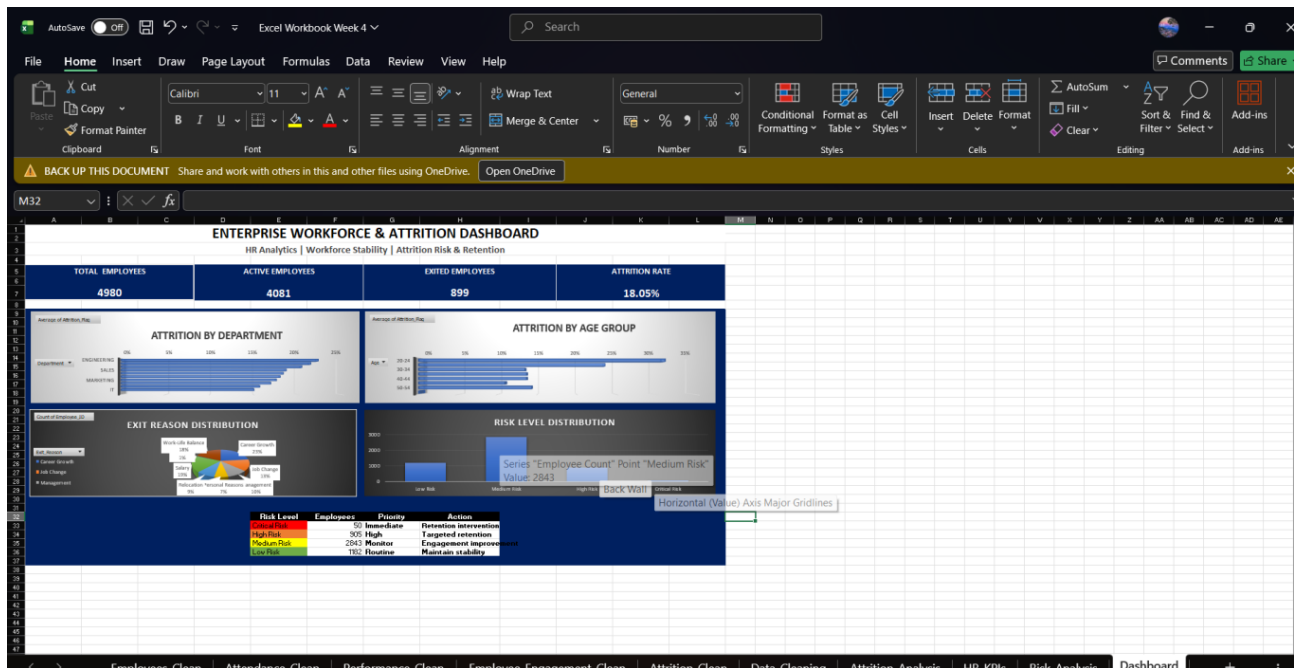


Figure 33. Excel dashboard evidence.

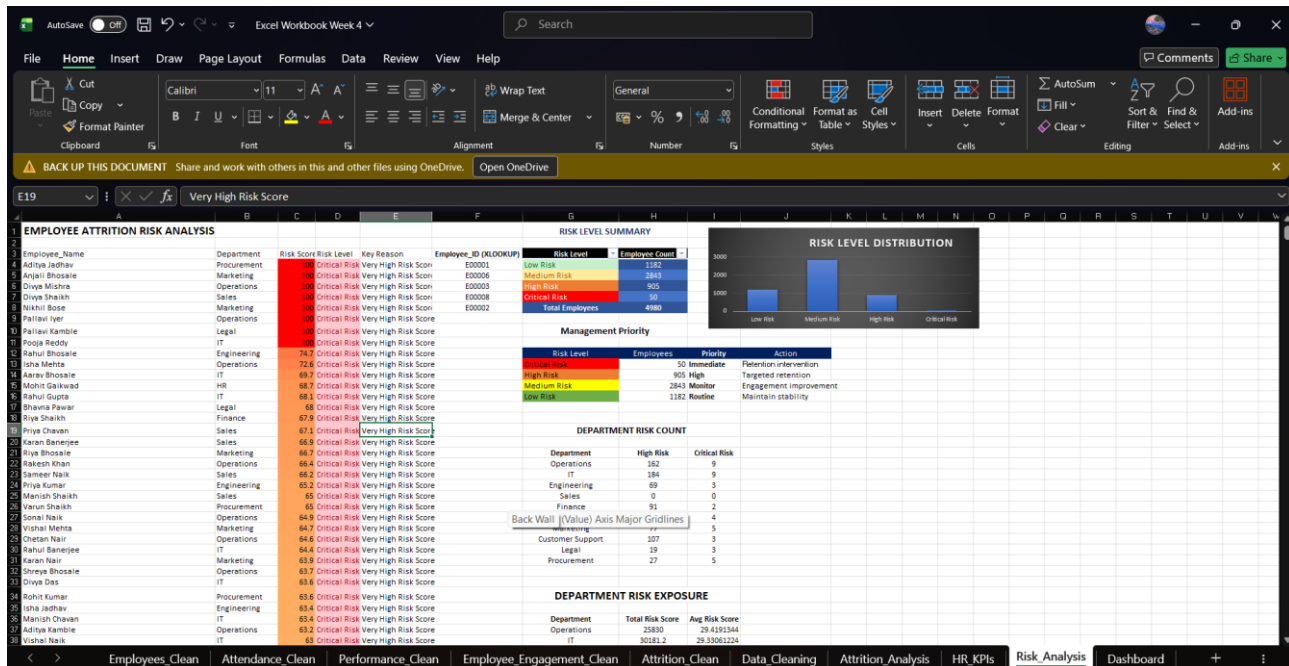


Figure 34. Refined Excel dashboard/analysis layout evidence.

26. SQL Analytics

SQL was used as a business analytics layer rather than as a theory-only exercise. The project requirement included JOIN, GROUP BY, HAVING, CASE, subqueries, CTE, window functions, ranking and date calculations. The report only labels the queries below as verified where they are explicitly present in the supplied project instructions.

EVIDENCE RULE

Where the complete SQL execution file is not available in the supplied screenshot package, the report does not claim execution beyond the verified query definitions.

26.1 Verified SQL Query 1 — Department Attrition

```
SELECT
    e.Department,
    COUNT(a.Employee_ID) * 1.0 / COUNT(e.Employee_ID) AS attrition_rate
FROM Employees e
LEFT JOIN Attrition a
    ON e.Employee_ID = a.Employee_ID
GROUP BY e.Department;
```

Element	Explanation
LEFT JOIN	Keeps the employee population while matching attrition where present.
COUNT	Counts employee/attrition records.
GROUP BY Department	Produces department-level output.
Calculation	Attrition count divided by employee count.
Business Question	Which departments have higher attrition?

26.2 Verified SQL Query 2 — Exit Reason Ranking

```
SELECT
    Exit_Reason,
    COUNT(*) AS exits,
    RANK() OVER (ORDER BY COUNT(*) DESC) AS reason_rank
```

```
FROM Attrition
GROUP BY Exit_Reason;
```

Element	Explanation
GROUP BY	Creates one row per exit reason.
COUNT	Measures exit volume.
RANK()	Ranks reasons from highest to lowest count.
Business Question	Which exit reasons dominate?

26.3 Verified SQL Query 3 — Tenure Bands with CTE

```
WITH tenure_banded AS (
  SELECT
    Employee_ID,
    CASE
      WHEN Tenure_Months <= 11 THEN '0-11'
      WHEN Tenure_Months <= 23 THEN '12-23'
      ELSE '24+'
    END AS tenure_band
  FROM Attrition
  WHERE Exit_Reason IS NOT NULL
    AND Exit_Type IS NOT NULL
    AND Tenure_Months IS NOT NULL
)
SELECT
  tenure_band,
  COUNT(*) AS exits
FROM tenure_banded
GROUP BY tenure_band;
```

26.4 Verified SQL Query 4 — Active Workforce Salary

```
SELECT
  Department,
  COUNT(*) AS Employee_Count,
  AVG(Salary) AS Avg_Salary
FROM Employees
WHERE Employment_Status = 'Active'
GROUP BY Department
ORDER BY Employee_Count DESC;
```

SQL Capability	Project Use
JOIN	Connect employee and attrition information.
GROUP BY	Department/reason aggregation.
CASE	Tenure banding.
RANK / Window Functions	Exit reason ranking.
CTE	Structured tenure transformation.
WHERE	Filter usable or active records.
ORDER BY	Sort analytical outputs.
Date Calculations	Support time/tenure analysis.

SQL DOCUMENTATION STANDARD

For each additional SQL query, document: business question → tables → logic → execution → result → validation → interpretation → business relevance. Do not claim a query was executed unless evidence exists.

27. Predictive Analytics

The predictive workflow prepared employee-level features, created an attrition target, accounted for class imbalance, split the data, trained models, generated predictions, tuned thresholds and evaluated precision, recall and F1.

Stage	What Was Done	Why It Matters
Feature preparation	Employee-level aggregated features	Prevents repeated event rows from dominating employee-level modelling.
Target creation	Attrition flag	Provides binary outcome.
Class imbalance	Balanced Logistic Regression / evaluation focus	Avoids relying only on majority-class accuracy.
Train/test split	Separate training and evaluation data	Measures generalisation.
Prediction	Generate class/probability outputs	Supports threshold analysis.
Threshold tuning	Compare thresholds	Align model with business cost of missed leavers.
Evaluation	Accuracy, precision, recall, F1, confusion matrix	Provides a more complete model view.

```
# =====
# BALANCED LOGISTIC REGRESSION:--->

balanced_model = LogisticRegression(
    class_weight="balanced",
    max_iter=1000,
    random_state=42
)

# Train balanced model
balanced_model.fit(
    X_train_scaled,
    y_train
)

# Predictions
balanced_y_pred = balanced_model.predict(
    X_test_scaled
)

print("Balanced Logistic Regression model trained successfully.")
```

Balanced Logistic Regression model trained successfully.

Figure 35. Logistic Regression implementation evidence.

Random Forest Confusion Matrix:

```
[[813  5]
 [ 61  2]]
```

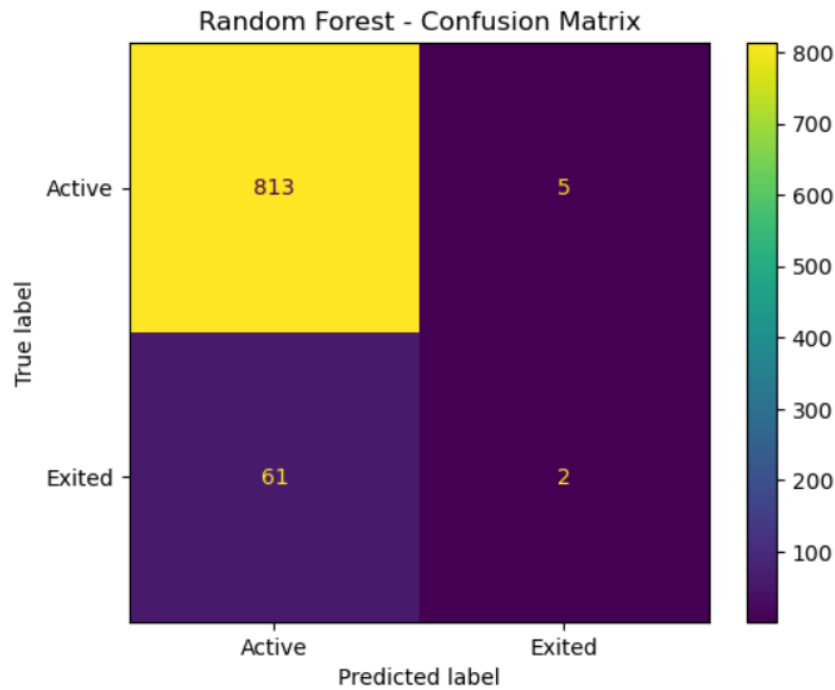


Figure 36. Random Forest model/confusion-matrix evidence.

28. Machine Learning Model Evaluation

28.1 Logistic Regression

Metric	Threshold 0.50	Final Threshold 0.70
Accuracy	72.19%	87.74%
Precision	17.27%	25.81%
Recall	76.19%	38.10%
F1	28.15%	30.77%

The threshold comparison shows the trade-off between precision and recall. The final selected threshold was 0.70, producing the highest listed F1 of 30.77% among the tested Logistic Regression thresholds.

Threshold	Precision	Recall	F1
0.30	11.29%	90.48%	20.07%
0.35	12.39%	87.30%	21.70%
0.40	13.70%	84.13%	23.56%
0.45	15.52%	82.54%	26.13%
0.50	17.27%	76.19%	28.15%
0.55	18.26%	63.49%	28.37%
0.60	20.69%	57.14%	30.38%
0.65	22.30%	49.21%	30.69%
0.70	25.81%	38.10%	30.77%

performance_classification

Performance Classification Finding

The workforce is primarily concentrated in the Meets Expectations category, with 2,189 employees, followed by 1,241 employees needing improvement and 1,113 high performers. A smaller group of 192 employees is classified as Outstanding, while 233 employees fall into the Critical category.

Performance, productivity, quality, manager rating, and target achievement show a consistent upward pattern across the five performance categories. Outstanding employees achieved an average target achievement of 106.24%, while Critical employees achieved only 71.91%.

This indicates a strong alignment between overall performance and the supporting performance indicators.

Figure 37. Logistic Regression classification report evidence.

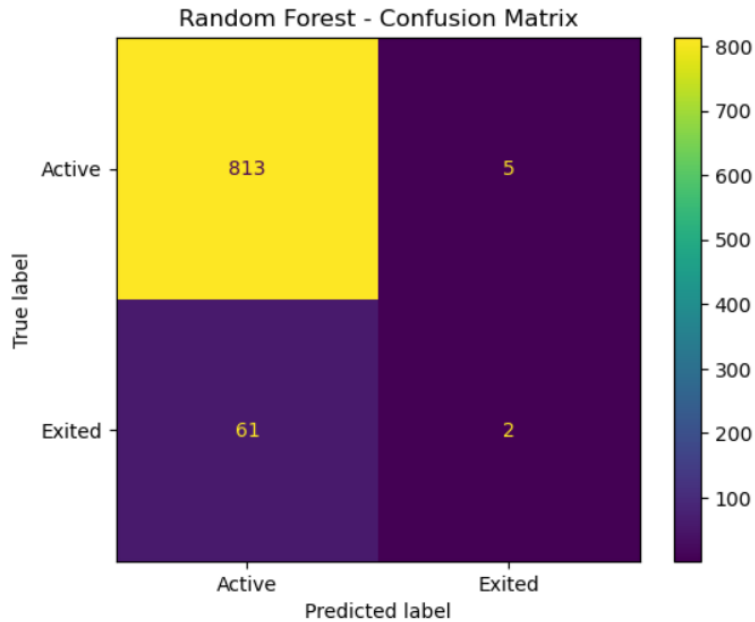


Figure 38. Logistic Regression confusion matrix table evidence.

Threshold Comparison

```
=====
Threshold: 0.30 | Precision: 11.29% | Recall: 90.48% | F1: 20.07%
Threshold: 0.35 | Precision: 12.39% | Recall: 87.30% | F1: 21.70%
Threshold: 0.40 | Precision: 13.70% | Recall: 84.13% | F1: 23.56%
Threshold: 0.45 | Precision: 15.52% | Recall: 82.54% | F1: 26.13%
Threshold: 0.50 | Precision: 17.27% | Recall: 76.19% | F1: 28.15%
Threshold: 0.55 | Precision: 18.26% | Recall: 63.49% | F1: 28.37%
Threshold: 0.60 | Precision: 20.69% | Recall: 57.14% | F1: 30.38%
Threshold: 0.65 | Precision: 22.30% | Recall: 49.21% | F1: 30.69%
Threshold: 0.70 | Precision: 25.81% | Recall: 38.10% | F1: 30.77%
```

Figure 39. Logistic Regression threshold comparison evidence.

28.2 Random Forest

Metric	Verified Value
Accuracy	92.51%
Precision	28.57%
Recall	3.17%
F1	5.71%

Threshold	Precision	Recall	F1
0.20	26.00%	20.63%	23.01%
0.25	35.48%	17.46%	23.40%
0.30	40.91%	14.29%	21.18%
0.35	37.50%	9.52%	15.19%
0.40	33.33%	6.35%	10.67%
0.45	37.50%	4.76%	8.45%

0.50	28.57%	3.17%	5.71%
------	--------	-------	-------

Random Forest has high accuracy but very low recall in the verified evaluation. For attrition use cases, missing actual leavers can be more damaging than accepting additional false positives. The model therefore should not be presented as production-ready.

model_comparison

	Model	Accuracy	Precision	Recall	F1 Score
0	Original Logistic Regression	92.96	66.67	3.17	6.06
1	Balanced Logistic Regression	72.19	17.27	76.19	28.15
2	Balanced Logistic Regression (Threshold 0.70)	87.74	25.81	38.10	30.77
3	Random Forest	92.51	28.57	3.17	5.71

Final Model Selection:--->

The Balanced Logistic Regression model with a decision threshold of 0.70 was selected as the final model among the tested configurations because it achieved the highest F1 Score of 30.77%.

The original Logistic Regression and Random Forest models achieved high accuracy but very low recall, indicating that they failed to identify most employees who actually exited.

Although the balanced model at the default threshold of 0.50 achieved higher recall (76.19%), the 0.70 threshold provided the highest F1 Score (30.77%) among the tested thresholds.

Figure 40. Model comparison evidence.

```
# BALANCED MODEL CONFUSION MATRIX:--->

balanced_cm = confusion_matrix(
    y_test,
    balanced_y_pred
)

print("Balanced Model Confusion Matrix:")
print(balanced_cm)
```

```
Balanced Model Confusion Matrix:
[[588 230]
 [ 15  48]]
```

Figure 41. Model confusion matrix evidence.

28.3 Model Comparison

Model	Threshold	Accuracy	Precision	Recall	F1	Business Usefulness
Logistic Regression	0.70	87.74%	25.81%	38.10%	30.77%	More balanced recall/F1 trade-off; useful for analytical early warning with review.
Random Forest	0.50	92.51%	28.57%	3.17%	5.71%	High accuracy but misses most actual leavers; not production-ready for retention screening.

MODEL LESSON

Accuracy alone is not enough for attrition prediction. Recall and F1 must be reviewed because the business objective is to

identify employees who may leave, not merely to maximise the majority-class accuracy.

29. Business Insights

FINDING	EVIDENCE	IMPACT	RECOMMENDATION
Overall attrition	18.05% across 4,980 employees.	Retention should be treated as a management priority.	Maintain a regular attrition review cadence.
Engineering attrition	Engineering = 22%.	Highest verified departmental attrition.	Run a department-specific retention diagnosis.
HR attrition	HR = 21%.	HR itself has elevated turnover.	Review HR workload, career and manager experience.
Finance attrition	Finance = 20%.	Finance also exceeds IT in verified attrition.	Assess role-specific retention drivers.
IT attrition	IT = 15%.	Lower than the other listed departments.	Use IT as a comparison group, not as proof of best practice.
Early-career attrition	Age 20–24 = 31%; 25–29 = 24%.	Younger employees represent a key retention segment.	Strengthen onboarding, mentoring and progression.
Lower salary band	Lowest salary band = 35%.	Compensation level is associated with higher attrition.	Review pay competitiveness and progression.
Career growth	206 exits.	Largest recorded exit reason.	Create visible career pathways and internal mobility.
Salary	173 exits.	Second-largest recorded exit reason.	Review compensation bands and progression.
Work-life balance	160 exits.	Third-largest recorded exit reason.	Review workload, scheduling and flexibility.
Top three reasons	539 / 890 ≈ 60.6%.	Three themes dominate detailed usable exit reasons.	Prioritise career, salary and work-life interventions.
Voluntary exits	732.	Voluntary exits dominate.	Focus on preventable retention levers.
First 24 months	388 exits; ≈43.6% of 890.	Large share occurs early in tenure.	Create 0–24 month retention programme.
High risk	905.	Large high-risk population.	Prioritise manager/HR review.
Critical risk	50.	Smallest but highest urgency group.	Use individual intervention.
High + Critical	955; ≈19.2% of workforce.	About one-fifth of workforce is in top two risk levels.	Create a prioritised risk-review queue.
Engagement	Engagement ↔ attrition = -0.262206.	Higher engagement is associated with lower attrition.	Monitor engagement dimensions alongside exits.
Performance/productivity	0.759140 internal correlation.	Performance and productivity move together strongly.	Use them jointly for productivity reporting, avoiding redundant interpretation.

30. Final Management Questions

Question	Final Answer / Evidence Boundary
What is current attrition?	18.05%.
Which department highest?	Engineering, 22%.
Which group most likely to leave?	The strongest observed segment is age 20–24 at 31%; risk scoring further identifies 955 High/Critical employees.
Does salary influence attrition?	Associated with attrition; causation not established.
Does overtime influence attrition?	Near-zero verified correlation with attrition: 0.005853.
Does engagement influence retention?	Engagement-related variables have negative associations with attrition; causation is not established.
Which departments highest productivity?	Not conclusively established from the available analysis.
Which departments attendance issues?	Not conclusively established from the available analysis.
Major exit reasons?	Career Growth, Salary, Work-Life Balance are the top three.
Which employees are high risk?	905 High Risk + 50 Critical Risk = 955.
What should management prioritise?	Career growth, lower salary-band review, early-career retention, department-specific plans, risk-based interventions, work-life balance and manager support.

31. Errors & Problem Resolution

PROBLEM	WHAT I WAS TRYING TO DO	ACTUAL ISSUE	WHY	DIAGNOSIS	FIX	FINAL RESULT	LEARNING
Python — df not defined	Creating correlation analysis dataframe.	NameError: name 'df' is not defined	The dataframe variable was not defined in the active notebook/session or a different variable name was in use.	Checked active dataframe names and execution order.	Used the correct defined dataframe / corrected variable reference.	Correlation workflow resumed.	Consistent naming and execution order matter.
Python — ConfusionMatrixDisplay not defined	Displaying model confusion matrix.	NameError: name 'ConfusionMatrixDisplay' is not defined	The class had not been imported.	Checked the import section.	Imported ConfusionMatrixDisplay from sklearn.metrics.	Confusion matrix display became available.	Required classes must be imported before use.
Python — plt not defined	Creating plots.	NameError: name 'plt' is not defined	matplotlib.pyplot was not available under the plt alias.	Checked active imports.	Imported matplotlib.pyplot as plt.	Plotting code could run.	Notebook state and imports must be checked.
Python — stale/undefined dataframe	Running later notebook cells.	State/undefined dataframe / execution-order issue	Earlier setup cells had not been executed consistently or variable state had changed.	Reviewed execution sequence and variable names.	Re-ran required setup / used the correct dataframe.	Downstream cells could execute.	Notebook execution order is part of reproducibility.
Python — Excel handled with read_csv	Loading an input file.	Input was Excel rather than CSV	The file format did not match the reader being used.	Checked the actual file extension/type.	Used the appropriate Excel-reading method.	Dataset loaded correctly.	Match the pandas reader to the file format.
Excel — AVERAGEIFS #DIV/0!	Average risk score by department.	#DIV/0! for Sales	No usable numerical matching values were available for the calculation.	Inspected matching records and calculation inputs.	Wrapped AVERAGEIFS with IFERROR.	Controlled output returned.	Error handling should be deliberate, not cosmetic.
Excel — PivotChart/PivotTable linkage	Remove a field from the chart only.	PivotTable changed too.	PivotChart and PivotTable share the same pivot structure.	Observed source table changing with chart field removal.	Used a static/copy summary table where independent chart structure was required.	Dashboard chart could be controlled independently.	Understand object dependencies before editing.
Excel — wrong PivotChart workflow	Insert a chart from the intended data.	Create PivotChart dialog appeared unexpectedly.	PivotChart command was selected instead of normal Chart workflow.	Observed the dialog and cancelled.	Used the correct normal Chart workflow when working from static summaries.	Chart creation proceeded as intended.	Choose chart type based on source structure.
Excel — duplicate Department slicer	Add final slicers.	Two Department slicers existed.	Slicer was created more than once.	Reviewed dashboard and Report Connections.	Deleted duplicate and verified final connections.	Final slicer layout was clean.	Check Report Connections and dashboard state.
Power BI → Excel — missing Employee_ID	Export employee risk details.	Employee_ID missing in export.	Exported detail did not include the required key.	Inspected exported columns.	Recovered Employee_ID with XLOOKUP using Name + Department.	Operational Excel risk table became usable.	Source exports should preserve stable keys.

32. Practical Click-by-Click Implementation Log

32.1 Python — Click / Execution Workflow

Step	Action	Expected	Actual / Learning
1	Open Jupyter Notebook	Notebook environment opens	Project notebook ready
2	Import required libraries	Imports execute without errors	Libraries available
3	Load dataset	Dataframe created	File path and format verified
4	Inspect shape and columns	Rows/columns and names visible	Schema understood
5	Run head/sample	Representative rows visible	Initial data understanding
6	Run describe	Numeric summary visible	Ranges/outliers considered
7	Check nulls	Null counts visible	Missing-value actions identified
8	Check duplicates	Duplicate count visible	Duplication addressed
9	Perform validation	Business-rule checks run	Invalid records identified
10	Clean data	Validated dataframe created	Downstream analysis protected
11	Aggregate repeated employee-level data	One row per employee analytical features	Correlation/model inputs stabilised
12	Perform EDA	Distributions and relationships	Patterns identified
13	Create correlation analysis	Employee-level heatmap	Linear associations quantified
14	Create risk score	0–100 risk output	Employees prioritised

15	Train Logistic Regression	Model fitted	Baseline classification
16	Evaluate	Metrics/confusion matrix	Performance assessed
17	Tune threshold	Threshold comparison	Business trade-off assessed
18	Train Random Forest	Second model fitted	Alternative model assessed
19	Compare models	Accuracy/precision/recall/F1 table	Model usefulness evaluated

32.2 SQL — Click by Click

Step	Action	Validation / Learning
1	Open SQL environment	Confirm database/schema is selected.
2	Select database/schema	Ensure expected tables exist.
3	Open query editor	Prepare the business question.
4	Write query	Use verified syntax and actual field names.
5	Execute	Review query result grid.
6	Validate output	Compare counts/rates with known project evidence where applicable.
7	Interpret	Translate rows into business meaning.

32.3 Excel — Click by Click

Step	Excel Action	Purpose
1	Open workbook	Start reporting workflow.
2	Open target sheet	Select analytical source.
3	Select source data	Define range/table.
4	Insert → PivotTable	Create summary analysis.
5	Select destination	Place PivotTable logically.
6	Set Rows	Choose business dimension.
7	Set Values	Choose Count/Sum/Average.
8	Change aggregation	Match business question.
9	Sort descending	Highlight largest categories.
10	Format	Improve readability.
11	Create chart	Visualise the summary.
12	Create slicer	Add interactive filtering.
13	Report Connections	Connect relevant PivotTables.
14	Test filtering	Validate interaction.

32.4 XLOOKUP — Click by Click

Step	Action	Check
1	Open Risk_Analysis	Confirm missing Employee_ID column.
2	Add Employee_ID column	Create destination field.
3	Enter XLOOKUP	Use Name + Department composite key.
4	Press Enter	Confirm returned ID.
5	Fill down	Populate records.
6	Spot-check	Verify selected IDs manually.
7	Check Not Found	Investigate unmatched rows.
8	Document limitation	Names + departments may not be unique.

32.5 COUNTIFS / SUMIFS / AVERAGEIFS

Function	Click-by-Click	Validation
COUNTIFS	Create Department × Risk table → add headers → enter formula → fill across/down.	Check counts against risk population.
SUMIFS	Create Department Risk Exposure → enter formula → fill down.	Check totals and departments.
AVERAGEIFS / IFERROR	Enter AVERAGEIFS → observe #DIV/0! → inspect matches → wrap with IFERROR → verify.	Confirm controlled result and document reason.

32.6 Conditional Formatting

Step	Excel Path	Result
------	------------	--------

1	Select Risk Level	Target range selected.
2	Home → Conditional Formatting → Highlight Cells Rules → Text That Contains	Create category rules.
3	Low Risk → Green	Low priority visually identified.
4	Medium Risk → Yellow	Preventive attention identified.
5	High Risk → Orange	Priority review identified.
6	Critical Risk → Red	Highest priority identified.
7	Select Risk Score → Home → Conditional Formatting → Color Scales	Continuous risk colour scale.
8	Verify	Check colours against values.

32.7 PivotTable / PivotChart / Slicer

Task	Click Path / Action	Learning
PivotTable	Insert → PivotTable → Existing Worksheet → Rows → Values → Count Employee_ID → Sort	Build summary from business question.
PivotChart	Select PivotTable → chart workflow; if independent visual is needed use static summary + normal Chart.	PivotChart is linked to PivotTable.
Slicer	Select PivotTable → PivotTable Analyze → Insert Slicer → Department + Age → OK → resize → Report Connections → test.	Slicers can control connected PivotTables.
Duplicate slicer fix	Review dashboard → Report Connections → delete duplicate → test.	Always verify final interactive state.

32.8 Dashboard — Click by Click

Step	Action	Outcome
1	Create Dashboard sheet	Blank reporting canvas.
2	Set useful column widths	Expanded A:L layout.
3	Create title/subtitle	Dashboard context.
4	Create KPI cards	4,980 / 4,081 / 899 / 18.05%.
5	Insert charts	Attrition/risk views.
6	Add slicers	Department and Age filters.
7	Add background/formatting	Corporate visual hierarchy.
8	Review initial layout	Initial dashboard was too small.
9	Rebuild to A:L	Improved space and readability.
10	Final review	Confirm calculations, alignment and filtering.

32.9 Power BI — Click by Click

Step	Action	Outcome
1	Get Data	Import cleaned tables.
2	Check relationships	Confirm Employee_ID relationships.
3	Open required report page	Select management view.
4	Create visuals	KPI/cards/charts/tables.
5	Format	Apply consistent dashboard hierarchy.
6	Create risk table	Employee-level risk detail.
7	Apply conditional formatting	Prioritise risk visually.
8	Build six pages	Overview, Attrition, Productivity, Engagement, Risk, Action Center.
9	Export risk data	Create operational CSV.
10	Open CSV in Excel	Continue operational workflow.
11	Validate	Check recovered Employee_ID and risk fields.

Power BI Dashboard Evidence — Six Management Pages



Figure 42. Page 1 — Executive HR Overview.



Figure 43. Page 2 — Attrition Analytics.

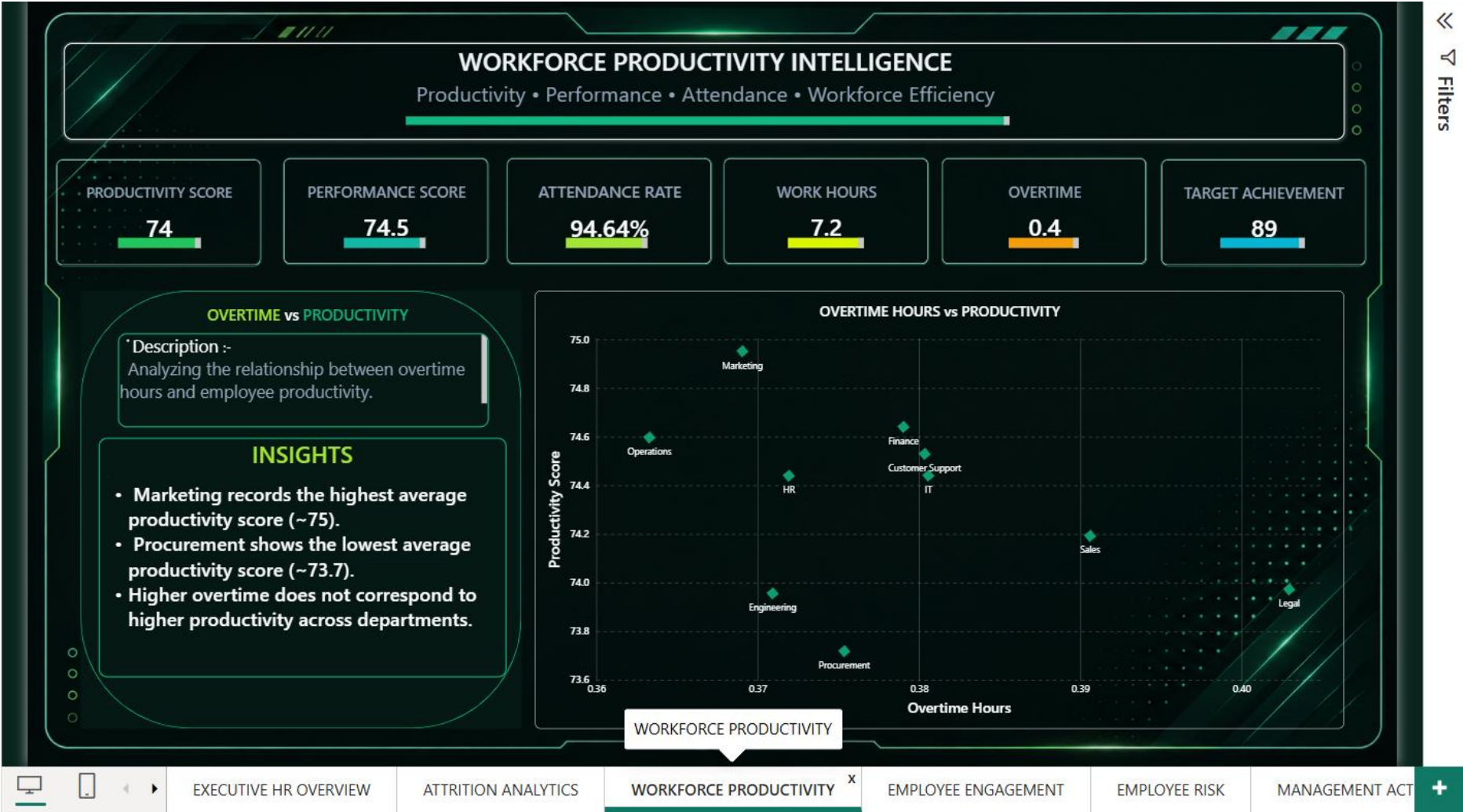


Figure 44. Page 3 — Workforce Productivity.



Figure 45. Page 4 — Employee Engagement.



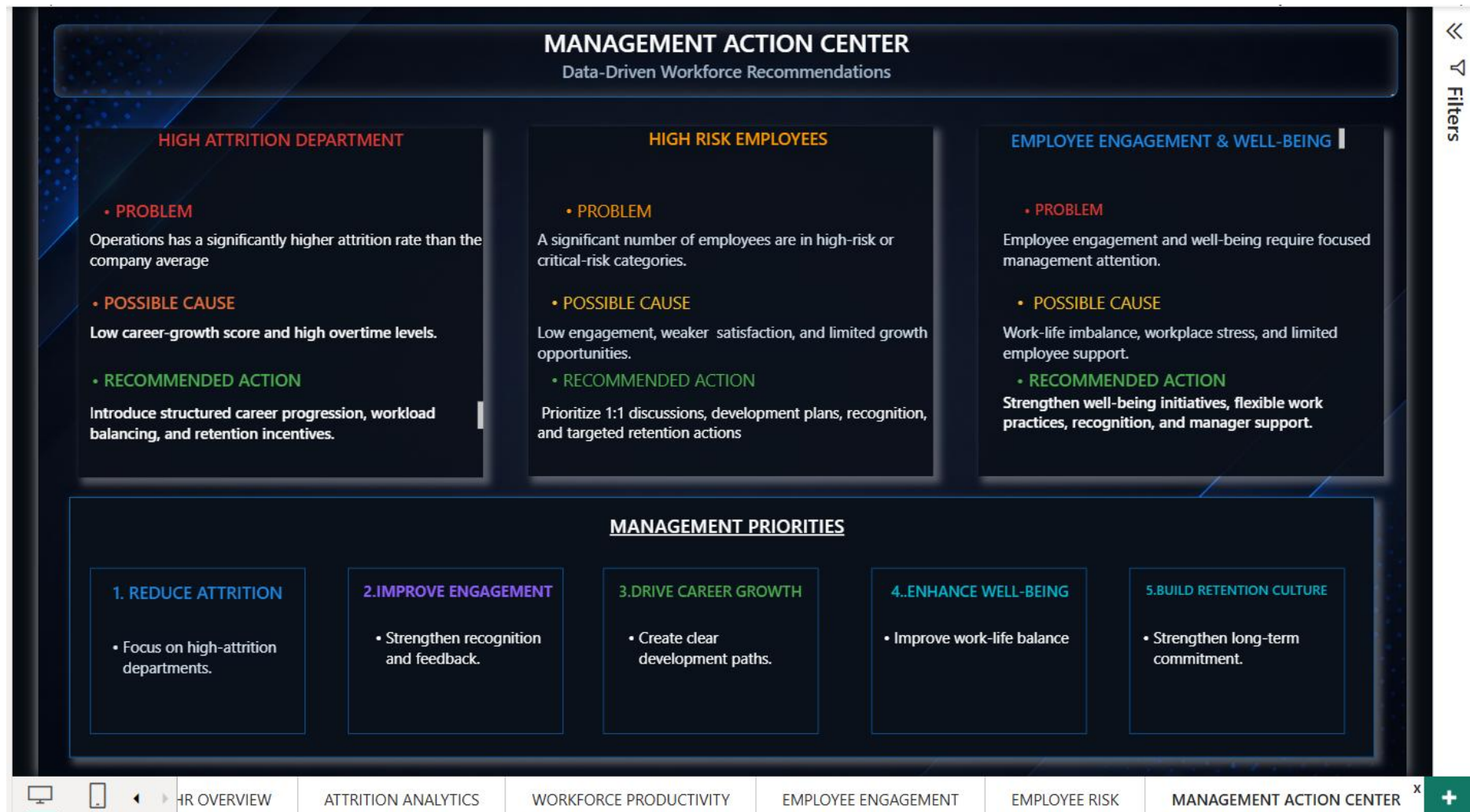


Figure 47. Page 6 — Management Action Center.

33. Project Learnings

- Data cleaning is broader than removing nulls: key integrity, ranges, dates, duplicates and business rules matter.
- Employee_ID consistency is central to combining HR datasets safely.
- Repeated Attendance, Performance and Engagement records need employee-level aggregation for employee-centric analysis.
- Notebook execution order and variable naming directly affect reproducibility.
- Debugging requires reading the actual error, checking the active state and then applying the smallest correct fix.
- Correlation is not causation; business recommendations should use cautious association language.
- Class imbalance means accuracy can be misleading in attrition prediction.
- Recall and F1 are important when the cost of missing a potential leaver is high.
- Threshold tuning is a business decision, not just a mathematical exercise.
- Excel formulas can create traceable operational outputs when source fields and logic are documented.
- PivotCharts are structurally linked to their PivotTables.
- Slicers need Report Connections validation to behave consistently.
- Power BI and Excel can form a reporting-to-operational workflow, but stable keys should ideally be preserved in exports.
- SQL translates management questions into reproducible aggregation and ranking logic.
- Python is useful for scalable profiling, feature engineering, correlation and modelling.
- Good analytics ends with business communication: findings, impact and recommended action.

34. Limitations

Limitation	Implication	How It Was Handled
899 vs 890 populations	Detailed exit fields have a smaller usable population.	Explained explicitly; not treated as contradiction.
Correlation does not prove causation	Observed relationships may have other drivers.	Use association language.
Risk score is not guaranteed prediction	A risk flag can produce false positives/negatives.	Use as early-warning framework.
Random Forest recall is low	Most actual leavers are missed by the verified model.	Do not call production-ready.
Some management questions lack evidence	Unsupported department productivity/attendance rankings should not be invented.	State not conclusively established.
Name + Department XLOOKUP	Composite key may not be unique.	Use stable Employee_ID in source systems where possible.
Power BI export omitted Employee_ID	Operational export needed key recovery.	Used XLOOKUP and documented limitation.
Data quality affects downstream analysis	Errors propagate into dashboards/models.	Validate before analysis.

35. MANAGEMENT RECOMMENDATIONS

FINDING	IMPACT	RECOMMENDATION	PRIORITY	RESPONSIBLE HR AREA
Strengthen Career Growth Opportunities	Career Growth is the largest recorded exit reason (206).	Career stagnation can create avoidable voluntary exits.	Create visible career paths, mentoring and internal mobility.	HIGH
Review Compensation at Lower Salary Levels	Lowest salary band shows 35% attrition; Salary accounts for 173 exits.	Compensation competitiveness is a clear retention signal.	Review pay bands, progression and market alignment.	HIGH
Focus on Early-Career Retention	20–24 attrition is 31%; 388 exits occurred in first 24 months.	Early tenure is a major retention window.	Introduce 30/60/90-day onboarding, mentoring and career checkpoints.	HIGH
Department-Specific Retention Plans	Engineering 22%, HR 21%, Finance 20%, IT 15%.	Retention drivers may differ by function.	Create targeted plans for higher-attrition departments.	HIGH
Prioritize High/Critical Risk Employees	955 employees are High/Critical Risk.	Large actionable review population.	Create a risk-prioritised HR review queue.	HIGH
Critical-Risk Individual Intervention	50 employees are Critical Risk.	Highest risk requires personalised action.	Conduct confidential stay interviews and manager review.	CRITICAL
Preventive Action for Medium-Risk Employees	2,843 employees are Medium Risk.	Prevention can be cheaper than late intervention.	Use engagement and career check-ins before risk escalates.	MEDIUM
Improve Work-Life Balance	160 exits cite Work-Life Balance.	Workload/flexibility may affect retention.	Review workload, scheduling, flexibility and	HIGH

Strengthen Manager Support	Manager Satisfaction correlation with attrition = - 0.225597.	Manager experience is an important engagement-related signal.	overtime patterns. Manager coaching and regular one-to-ones.	HIGH
Improve Engagement Monitoring	Engagement Score correlation = -0.262206.	Engagement is the strongest listed attrition correlation.	Track engagement trends with attrition monthly.	HIGH
Improve Exit Interview Data Quality	Unknown exit reasons = 5 in detailed records.	Better reason data improves actionability.	Standardise exit reason categories and mandatory capture.	MEDIUM
Improve Power BI → Excel Data Pipeline	Employee_ID was missing from exported risk detail.	Manual key recovery adds risk and effort.	Preserve stable employee keys in approved exports.	HIGH
Evaluate Models Using Recall and F1	Random Forest recall = 3.17%.	High accuracy can conceal missed leavers.	Use recall/F1 alongside accuracy in model governance.	HIGH
Regular Attrition Monitoring	Attrition patterns can change over time.	Static reports can become outdated.	Create monthly/quarterly attrition monitoring.	MEDIUM
Move from Reactive to Proactive Retention	955 employees are High/Critical Risk.	Risk analytics can support earlier intervention.	Combine risk, engagement and manager action plans.	HIGH

35.1 MANAGEMENT ACTION MATRIX

Finding	Business Impact	Recommended Action	Priority	Responsible HR Area
Strengthen Career Growth Opportunities	Career Growth is the largest recorded exit reason (206).	Career stagnation can create avoidable voluntary exits.	Create visible career paths, mentoring and internal mobility.	Learning & Development / HRBP
Review Compensation at Lower Salary Levels	Lowest salary band shows 35% attrition; Salary accounts for 173 exits.	Compensation competitiveness is a clear retention signal.	Review pay bands, progression and market alignment.	Compensation & Benefits
Focus on Early-Career Retention	20–24 attrition is 31%; 388 exits occurred in first 24 months.	Early tenure is a major retention window.	Introduce 30/60/90-day onboarding, mentoring and career checkpoints.	HRBP / Managers
Department-Specific Retention Plans	Engineering 22%, HR 21%, Finance 20%, IT 15%.	Retention drivers may differ by function.	Create targeted plans for higher-attrition departments.	HRBP / Department Heads
Prioritize High/Critical Risk Employees	955 employees are High/Critical Risk.	Large actionable review population.	Create a risk-prioritised HR review queue.	HR Analytics
Critical-Risk Individual Intervention	50 employees are Critical Risk.	Highest risk requires personalised action.	Conduct confidential stay interviews and manager review.	HRBP / Managers
Preventive Action for Medium-Risk Employees	2,843 employees are Medium Risk.	Prevention can be cheaper than late intervention.	Use engagement and career check-ins before risk escalates.	HRBP / Managers
Improve Work-Life Balance	160 exits cite Work-Life Balance.	Workload/flexibility may affect retention.	Review workload, scheduling, flexibility and overtime patterns.	HR / Operations
Strengthen Manager Support	Manager Satisfaction correlation with attrition = - 0.225597.	Manager experience is an important engagement-related signal.	Manager coaching and regular one-to-ones.	People Managers
Improve Engagement Monitoring	Engagement Score correlation = -0.262206.	Engagement is the strongest listed attrition correlation.	Track engagement trends with attrition monthly.	HR Analytics
Improve Exit Interview Data Quality	Unknown exit reasons = 5 in detailed records.	Better reason data improves actionability.	Standardise exit reason categories and mandatory capture.	HR Operations
Improve Power BI → Excel Data Pipeline	Employee_ID was missing from exported risk detail.	Manual key recovery adds risk and effort.	Preserve stable employee keys in approved exports.	HR Analytics / BI
Evaluate Models Using Recall and F1	Random Forest recall = 3.17%.	High accuracy can conceal missed leavers.	Use recall/F1 alongside accuracy in model governance.	Analytics
Regular Attrition Monitoring	Attrition patterns can change over time.	Static reports can become outdated.	Create monthly/quarterly attrition monitoring.	HR Analytics
Move from Reactive to Proactive Retention	955 employees are High/Critical Risk.	Risk analytics can support earlier intervention.	Combine risk, engagement and manager action plans.	HR Leadership

36. Conclusion

The Enterprise Workforce Productivity & Employee Attrition Analytics System demonstrates an end-to-end Data Analyst Intern workflow: data was profiled, cleaned and validated; repeated records were aggregated to employee level; attrition, salary, attendance, productivity, performance and engagement were analysed; correlations and risk scores were created; predictive models were evaluated; SQL and Excel were used for business analysis; and Power BI was used for interactive management reporting.

The project identifies an overall attrition rate of 18.05%, 899 workforce-level exits and 955 High/Critical Risk employees. The detailed usable attrition population contains 890 records for exit reason, exit type and tenure analysis. Career Growth, Salary and Work-Life Balance are the three leading recorded reasons, while early-career employees and the lower salary band show higher observed attrition.

The modelling stage also provides an important analytical lesson: high accuracy is not sufficient for an attrition model. Random Forest reached 92.51% accuracy but only 3.17% recall, while the selected Logistic Regression threshold of 0.70 produced 87.74% accuracy, 25.81% precision, 38.10% recall and 30.77% F1. These results support cautious use of predictive analytics as an early-warning aid rather than an automated decision engine.

PROJECT IMPACT SUMMARY

4,980 Employees Analysed • 899 Exited • 18.05% Overall Attrition • 955 High/Critical Risk • 5 HR Datasets • Python + SQL + Excel + Power BI • 15+ Business Insights • Management Recommendations

37. Appendix

Appendix A — Python Code Reference

```
# Core profiling
df.head()
df.sample()
df.shape
df.columns
df.describe()
df.isnull().sum()
df.duplicated().sum()
df.value_counts()

# Attrition flag
df["Attrition_Flag"] = df["Employment_Status"].apply(
    lambda x: 1 if x == "Exited" else 0
)
```

The code block above is a compact reference of the documented profiling and flagging operations. It is not presented as a complete copy of the project notebook.

Appendix B — SQL Query Reference

```
-- Department-wise attrition
SELECT
    e.Department,
    COUNT(a.Employee_ID) * 1.0 / COUNT(e.Employee_ID) AS attrition_rate
FROM Employees e
LEFT JOIN Attrition a
    ON e.Employee_ID = a.Employee_ID
GROUP BY e.Department;
```

```
-- Exit reason ranking
SELECT
    Exit_Reason,
    COUNT(*) AS exits,
    RANK() OVER (ORDER BY COUNT(*) DESC) AS reason_rank
FROM Attrition
GROUP BY Exit_Reason;
```

```
-- Tenure banding with CTE
WITH tenure_banded AS (
    SELECT Employee_ID,
           CASE
               WHEN Tenure_Months <= 11 THEN '0-11'
               WHEN Tenure_Months <= 23 THEN '12-23'
               ELSE '24+'
           END AS tenure_band
    FROM Attrition
    WHERE Exit_Reason IS NOT NULL
          AND Exit_Type IS NOT NULL
          AND Tenure_Months IS NOT NULL
)
```

```
SELECT tenure_band, COUNT(*) AS exits
FROM tenure_banded
GROUP BY tenure_band;
```

Appendix C — Excel Formula Reference

Formula	Purpose
=XLOOKUP(A4&B4,Employees_Clean!B:B&Employees_Clean!E:E,Employees_Clean!A:A,"Not Found")	Recover Employee_ID from Name + Department.
=COUNTIFS(\$B:\$B,G22,\$D:\$D,H\$21)	Department × Risk Level count.
=SUMIFS(\$C:\$C,\$B:\$B,G37)	Total Risk Score by Department.
=IFERROR(AVERAGEIFS(\$C:\$C,\$B:\$B,G37),0)	Average Risk Score by Department with error handling.
=IF(Employee_Master!N2="Exited",1,0)	Binary attrition flag.

Appendix D — PivotTable Evidence

The supplied screenshot package includes Excel evidence for attrition analysis, COUNTIFS, SUMIFS, XLOOKUP, cleaning and dashboard construction. The screenshots are integrated into the relevant chapters rather than repeated here.

Appendix E — PivotChart Evidence

PivotChart linkage and the static-summary workaround are documented in Chapter 22 and the error chapter.

Appendix F — Slicer Evidence

Department and Age slicer creation, connections and duplicate-slicer resolution are documented in Chapter 24.

Appendix G — Dashboard Evidence

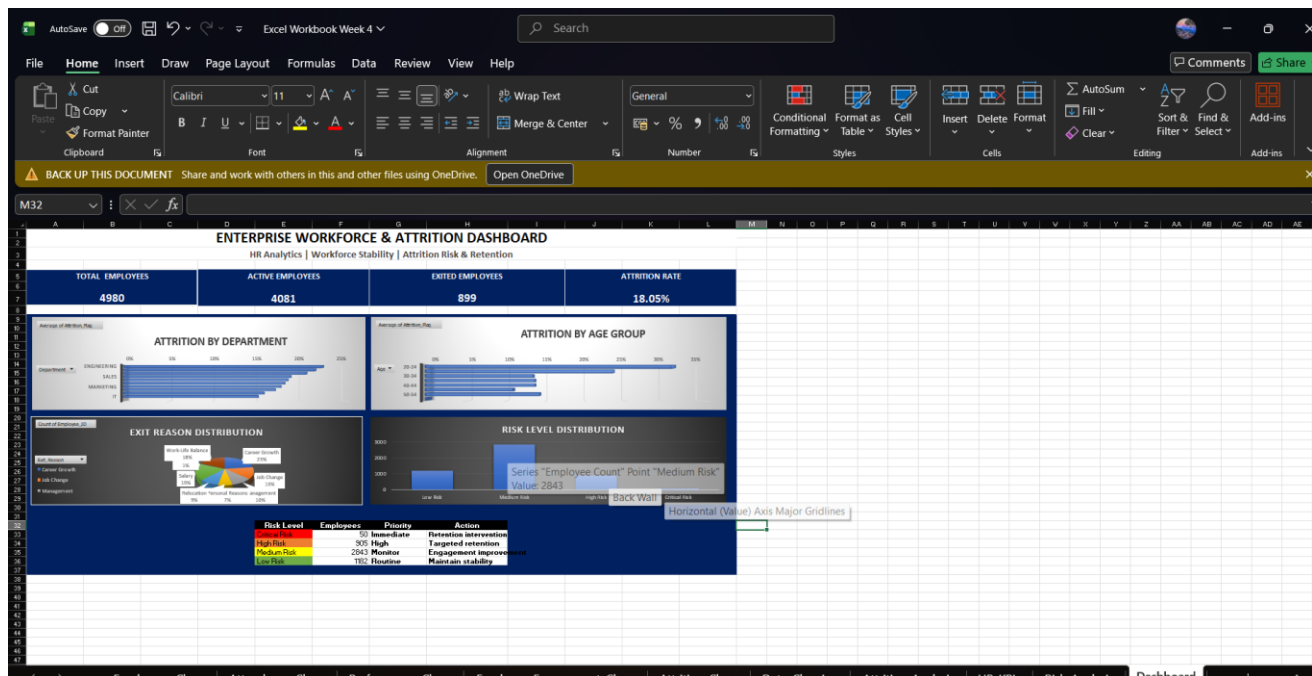


Figure 48. Excel dashboard evidence (appendix reference).

Appendix H — Power BI Evidence

All six Power BI pages are shown together in the landscape evidence section, Figures 42–47.

Appendix I — Error Evidence

```
# =====
# BALANCED LOGISTIC REGRESSION:--->

balanced_model = LogisticRegression(
    class_weight="balanced",
    max_iter=1000,
    random_state=42
)

# Train balanced model
balanced_model.fit(
    X_train_scaled,
    y_train
)

# Predictions
balanced_y_pred = balanced_model.predict(
    X_test_scaled
)

print("Balanced Logistic Regression model trained successfully.")
```

Balanced Logistic Regression model trained successfully.

Figure 49. Python execution/model evidence supporting the documented workflow.

Random Forest Confusion Matrix:

```
[[813  5]
 [ 61  2]]
```

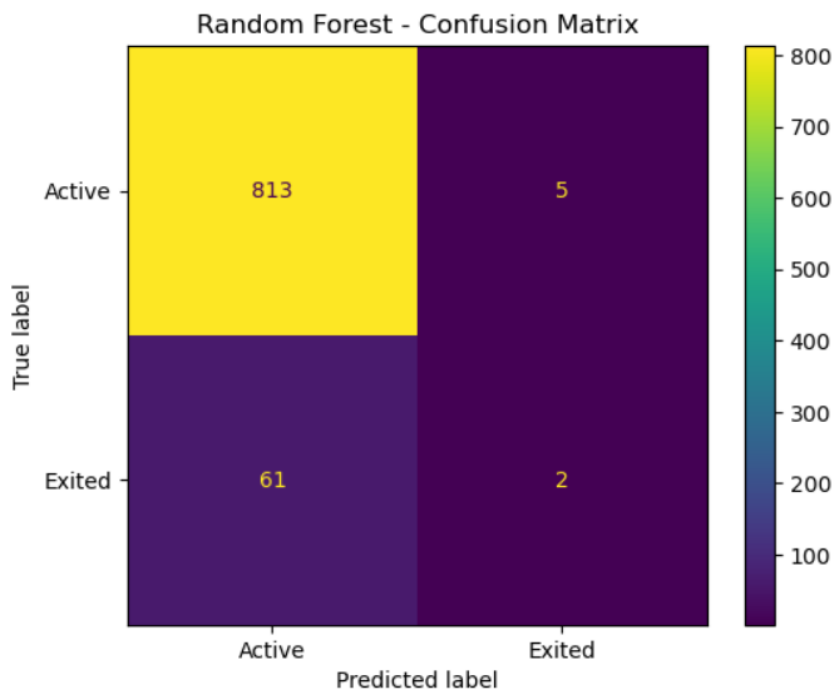


Figure 50. Random Forest/confusion-matrix evidence.

Appendix J — Correlation Values

Metric	Value
Salary ↔ Attrition	-0.092928
Experience_Years ↔ Attrition	-0.110673
Avg_Work_Hours ↔ Attrition	0.018315
Avg_Overtime_Hours ↔ Attrition	0.005853
Avg_Performance_Score ↔ Attrition	-0.003718
Avg_Productivity_Score ↔ Attrition	0.004357
Avg_Job_Satisfaction ↔ Attrition	-0.223238
Avg_Work_Life_Balance ↔ Attrition	-0.239754
Avg_Manager_Satisfaction ↔ Attrition	-0.225597
Avg_Career_Growth ↔ Attrition	-0.207363
Avg_Engagement_Score ↔ Attrition	-0.262206

Appendix K — Machine Learning Metrics

Model	Accuracy	Precision	Recall	F1
Logistic Regression @ 0.70	87.74%	25.81%	38.10%	30.77%
Random Forest @ 0.50	92.51%	28.57%	3.17%	5.71%

Appendix L — Click-by-Click Implementation Log

The full implementation log is provided in Chapter 32, including Python execution, SQL execution, Excel formulas, PivotTables, PivotCharts, slicers, dashboard construction and Power BI export.

Appendix M — Screenshot Index

Figure	Screenshot	Description	Chapter
1	Employees_Clean.png	Employee master evidence	Data Architecture
2	Employees_Clean.png	Employees_Clean worksheet	Data Cleaning
3	Attendance_Clean.png	Attendance_Clean worksheet	Data Cleaning
4	Performance_Clean.png	Performance_Clean worksheet	Data Cleaning
5	Employee_Engagement_Clean.png	Employee_Engagement_Clean worksheet	Data Cleaning
6	Attrition_Clean.png	Attrition_Clean worksheet	Data Cleaning
7	Missing Values & Duplicate.png	Missing values and duplicate validation	Data Cleaning
8	Validation.png	Validation checks	Data Cleaning
9	Data_Cleaning.png	Data cleaning worksheet	Data Cleaning
10	python libraries.png	Python libraries/imports	EDA
11	loading data in python.png	Loading data in Python	EDA
12	Data inspection python.png	Python data inspection	EDA
13	HR WORKFORCE INTEGENCE.png	Executive HR Overview	Workforce Overview
14	ATTRITION ANALYTICS.png	Attrition Analytics	Attrition
15	Attrition_Analysis.png	Attrition worksheet	Attrition
16	WORKFORCE PRODUCTIVITY.png	Workforce Productivity	Productivity
17	EMPLPOYEE ENGAGEMENT.png	Employee Engagement	Engagement
18	Corelation matrix heatmap.png	Correlation heatmap	Correlation
19	Risk scoring.png	Risk scoring code/output	Risk Scoring
20	EMPLOYEE RISK.png	Employee Risk dashboard	Risk Scoring
21	Xlookup.png	Risk export/XLOOKUP	Power BI → Excel
22	Dashboard.png	Excel dashboard	Excel Workbook
23	Xlookup.png	XLOOKUP formula	Excel Formula
24	Countifs.png	COUNTIFS table	Excel Formula
25	Sumif.png	SUMIFS table	Excel Formula
26	Screenshot 2026-08-31 125143.png	Department risk calculation	Excel Formula
27	Data_Cleaning.png	Attrition flag/calculation area	Excel Formula
28	Attrition_Analysis.png	Pivot/attrition analysis	PivotTables
29	Sumif.png	Summary calculation evidence	PivotTables
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31	Xlookup.png	Conditional formatting/risk table	Conditional Formatting
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33	Dashboard.png	Excel dashboard	Excel Dashboard
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35	Logistic Regression py.png	Logistic Regression implementation	Predictive Analytics
36	Random Forest.png	Random Forest evidence	Predictive Analytics
37	Classification Report.png	Classification report	ML Evaluation
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40	MOdel Comparison.png	Model comparison	ML Evaluation
41	Confusion matrix.png	Confusion matrix	ML Evaluation
42	HR WORKFORCE INTEGENCE.png	Power BI Page 1	Power BI
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50	Random Forest.png	Random Forest evidence	Appendix